A Project Report on

VayuNetra: AI-Powered Carbon Footprint and Emission Intelligence Platform

Submitted in partial fulfillment of the requirements for the award
of the degree of

Bachelor of Engineering

in

Computer Science and Engineering (Data Science)

by

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Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, We have adequately cited and referenced the original sources. We also declare that We have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

VayuNetra is an AI-driven platform focused on environmental sustainability, helping people and communities track, study, and cut down on carbon emissions, especially those linked to transportation. It works in the area of AI for environmental monitoring by using rule-based models for calculating CO2 emissions, natural language processing, and location data to figure out personal carbon footprints and offer practical tips for eco-friendly living. The platform supports sustainable behavior through a digital system that rewards users with carbon credits, manages user roles, and provides interactive dashboards to show progress at both individual and community levels. By combining real-time environmental data with AI predictions about pollution spikes, VayuNetra helps users make smarter decisions that reduce their environmental impact. The system evaluates multiple parameters, including travel mode, distance, fuel type, vehicle age, etc. Based on these insights, VayuNetra provides real-time, AI-generated sustainability tips that help users adopt greener travel choices and lifestyle habits. The platform features a digital carbon credit system that rewards employees and employers for eco-friendly actions and reductions in emissions.The system issues digital carbon credits only to employees and employers, and uses role-based access for employees, employers, banks, and admins. Its goal is to increase awareness, foster teamwork, and create real change toward a cleaner, more sustainable world.

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List of Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
CO_2	Carbon Dioxide
EF	Emission Factor
GHG	Greenhouse Gas
DB	Database
UI	User Interface
UX	User Experience
SQL	Structured Query Language
ML	Machine Learning
COE	Carbon Offset Exchange
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JSON	JavaScript Object Notation

Chapter 1

Introduction

VayuNetra is an AI-powered, web-based platform that enables individuals and communities to track, analyze, and reduce transportation-related carbon emissions. At its core, the system uses a rule-based CO_2 emission model combined with geolocation to compute personalized footprints from real travel activity. Users enter or log trip details, and VayuNetra factors in travel mode, distance, fuel type, vehicle condition/age, occupancy, frequency, and proximity to major industrial zones to estimate emissions with greater contextual accuracy. These results are translated into clear metrics and visual dashboards, helping users understand their impact over time at both the individual and community levels.

Building on measurement, VayuNetra provides AI-driven insights—including natural-language suggestions and real-time tips—to nudge greener choices such as public transport, carpooling, cycling, or optimized trip planning. The platform strengthens motivation through a digital carbon-credit system, where employees and employers earn and view credits in secure wallets for eco-friendly behavior, while role-based access (employees, employers, banks, admins) supports governance, reporting, and program oversight.

Beyond just tracking emissions, VayuNetra encourages sustainable habits through a digital carbon credit system, where users earn rewards for eco-friendly travel choices like carpooling, cycling, or using public transport. Interactive dashboards and reports display both individual and community-level progress, fostering awareness and collaboration toward sustainable mobility. Ultimately, VayuNetra aims to make carbon tracking simple, transparent, and motivating, contributing to a cleaner and more sustainable environment.

The system's backend is developed using the Django framework, with SQLite as the database for storing user data, emission factors, trip logs, and credit transactions. The frontend is built using HTML and Tailwind CSS to provide an interactive, user-friendly interface that works seamlessly across devices. The entire platform is hosted on Render Cloud Services to ensure scalability, reliability, and real-time accessibility. Each user type—employee, employer, bank, or administrator—has a distinct dashboard, designed to deliver personalized data visibility and analytics based on their respective roles. This modular, role-based architecture ensures efficient management of emissions data and facilitates transparent oversight of the carbon credit system.

In addition to emission computation and visualization, VayuNetra integrates location intelligence to analyze users' proximity to high-pollution or industrial zones. This contextual awareness allows the platform to correlate travel emissions with localized air quality, providing users with dynamic feedback on how their activities contribute to or offset carbon output in specific environments. Using AI-based prediction models, the system can forecast

pollution surges and suggest alternative travel routes or timing for reduced impact, thereby making sustainability not only achievable but also practical in daily commuting patterns.

VayuNetra's design philosophy centers around awareness, motivation, and collaboration. The platform transforms complex carbon emission calculations into intuitive visuals, giving users a clear sense of accountability and progress. It bridges the gap between information and action by merging environmental data with behavioral nudges and reward mechanisms. Over time, this helps users internalize sustainability as a daily routine rather than an abstract concept. The integration of community-based tracking further amplifies the collective effort by showing how small individual changes contribute to broader environmental goals, reinforcing the belief that sustainable change is achievable when individuals act together.

Ultimately, VayuNetra represents a step forward in applying artificial intelligence for environmental good. It leverages technology not only for data analysis but also for social impact—making environmental responsibility measurable, engaging, and rewarding. By combining AI-driven analytics, real-time geolocation data, and a transparent incentive model, the system aspires to create a culture of sustainability and conscious mobility that aligns with global efforts to reduce carbon emissions and combat climate change.

1.1 Motivation

We were encouraged to do this project because transportation is one of the biggest contributors to pollution and climate change, and most people are unaware of their personal impact.

Transportation is a dominant source of carbon emissions and a key driver of air pollution and climate change, yet individuals rarely perceive or quantify the impact of their everyday travel. Existing mechanisms such as PUC certifications provide only periodic compliance checks and do not translate real journeys, fuel choices, or vehicle condition into a continuous, comprehensible carbon footprint. This awareness gap limits informed decision-making and weakens the adoption of low-emission behaviors. VayuNetra is motivated by the need to turn raw travel activity into clear, personalized insight and to convert that insight into sustained action. By estimating a user's footprint from factors like mode, distance, fuel type, vehicle condition, and occupancy—and then reflecting it back through understandable metrics, suggestions, and progress views—the system seeks to build environmental literacy at the individual and community level. Pairing measurement with a carbon-credit mechanism provides a tangible incentive structure, aligning personal benefits with collective climate goals and addressing the current absence of continuous feedback and meaningful motivation to choose greener options. Our goal is to create a system that makes people aware of their emissions and motivates them to adopt greener choices in daily life.

1.2 Problem Statement

The rapid growth of transportation has led to rising carbon emissions, causing climate change, pollution, and health risks. Existing systems like PUC only give periodic checks and fail to provide continuous, actionable insights into vehicle emissions. There is also no

transparent system to motivate people to adopt greener travel. VayuNetra fills this gap with an AI-powered platform that monitors carbon footprints and rewards eco-friendly practices.

Transportation contributes a major share of global greenhouse gas emissions due to the continuous use of fossil fuels in vehicles. However, most individuals lack awareness of how much their everyday travel—such as short commutes, carpooling choices, or fuel selection—impacts the environment. Current government-led mechanisms such as Pollution Under Control (PUC) certificates only test emissions at fixed intervals and provide no personalized or real-time data. As a result, vehicle owners and organizations are unable to track their cumulative environmental impact or understand how small behavioral changes could lead to significant emission reductions.

Moreover, existing emission estimation systems are either corporate-focused or lack transparency and accessibility for individual users. There is no unified digital framework that integrates data from multiple sources—such as distance traveled, vehicle condition, or local pollution levels—to offer contextual insights. This limitation prevents users from understanding their real carbon footprint and taking informed action to reduce it. Without regular monitoring and relatable feedback, awareness remains low, and the adoption of eco-friendly alternatives such as carpooling, public transport, or electric vehicles continues to be limited.

Another major issue lies in the absence of an incentive-based mechanism to encourage and sustain behavioral change. Even when individuals are aware of environmental impacts, they often lack motivation to consistently choose greener options because there is no tangible reward system linked to sustainability. There is also a lack of data-driven transparency that allows organizations, banks, and government bodies to evaluate or support carbon-reduction initiatives. Therefore, there is an urgent need for a comprehensive, AI-driven system that continuously tracks emissions, offers meaningful feedback, and motivates users through measurable, reward-based mechanisms.

VayuNetra addresses this issue by integrating rule-based emission calculations, GPS-based contextual tracking, and AI-generated recommendations into one cohesive platform. It fills the existing gap between awareness and action by making emission tracking real-time, transparent, and personally engaging. Through carbon credit rewards and interactive dashboards, VayuNetra transforms sustainability from a compliance activity into a daily, data-driven habit that empowers individuals and organizations to actively reduce their carbon footprint.

1.3 Objectives

VayuNetra seeks to transform sustainability in everyday mobility by integrating intelligent, data-driven tools that make carbon tracking accurate, practical, and engaging. This approach focuses on empowering individuals and communities to understand their transportation-related emissions, receive actionable guidance, and adopt greener habits through transparent feedback and incentives. To achieve this vision, the following objectives have been established:

- Use a rule-based model to calculate each user's carbon footprint, offer realtime sustainability tips using AI and natural language processing, and reward eco-friendly travel through a digital carbon credit system.

- Manage users with different access levels for employees, employers, banks, and admins; store and display carbon credits in secure digital wallets; and encourage environmental awareness with emission trends, rewards based on behavior, and community progress dashboards.

- Track group impact by collecting user data into community goals, showing progress, and using AI trained on travel and activity patterns to predict future emissions.

- To determine where the user is, you can use GPS or IP location tracking. Check if they are close to major industrial zones. Use pollution data from reliable sources or fixed datasets to assess carbon monoxide levels, particularly during nighttime industrial operations.

- Offer personalized insights by comparing the user's carbon savings with the emissions from nearby industries. Use emotional messages and visuals to highlight their impact and using AI to predict potential pollution increases.

1.4 Scope

The scope of VayuNetra encompasses the design and implementation of an intelligent, AI-powered platform that enables users to understand and reduce transportation-based carbon emissions. The system integrates data analytics, artificial intelligence, and real-time environmental monitoring to provide meaningful insights and promote behavioral change toward sustainable mobility. The detailed scope of the project is explained below:

1. Development of an AI-powered platform for calculating and analyzing vehicle carbon footprints using fuel consumption and travel data:
   At the core of VayuNetra lies a rule-based CO_2 emission model designed to accurately estimate individual and organizational carbon footprints. The platform processes multiple travel-related parameters such as mode of transport, distance traveled, vehicle type, age, fuel type, and occupancy rate. It also incorporates contextual factors like proximity to industrial zones and frequency of travel. Using these parameters, the system calculates emissions in kilograms of CO_2 based on standardized emission factors. This model provides a transparent and scientifically grounded way to quantify daily carbon output, allowing users to monitor the impact of their transportation habits over time. The emission logic is supported by a Django-based backend that processes and stores all calculations securely in an SQLite database.

2. Provision of actionable insights and reports for individual users, employers, banks, and administrators:
   VayuNetra is structured as a multi-role system, where each user category has distinct access privileges and data perspectives. Individual users receive personalized insights about their daily or weekly travel emissions, while employers can view the aggregated emission data of their employees to measure workplace sustainability performance.

Banks and administrators have the ability to view summarized reports, analyze credit transactions, and track environmental progress at a macro level. The system generates automated reports and trend analyses that help users understand variations in their emission patterns. These insights are presented through AI-generated natural language summaries and graphical dashboards that convert technical data into simple, comprehensible information, ensuring inclusivity and ease of use.

3. Design and implementation of a Carbon Credit System that rewards users for adopting eco-friendly driving practices and low-emission vehicles:
   One of the most innovative aspects of VayuNetra is its integrated carbon credit reward mechanism. This feature translates a user's emission reduction efforts into measurable digital credits, creating a gamified yet practical incentive to adopt sustainable practices. Employees and employers can earn credits for actions such as choosing public transport, cycling, carpooling, or using electric vehicles. These credits are stored in secure digital wallets managed within the platform, ensuring data integrity and traceability. Employers can use these credits to promote workplace sustainability programs, while banks and administrators oversee the distribution and validation of rewards to maintain fairness and transparency. This mechanism bridges the gap between environmental awareness and actual behavioral change by providing tangible benefits for green actions.

4. Creation of an Emission Intelligence Dashboard for clear and meaningful visualization of carbon footprint data:
   The Emission Intelligence Dashboard acts as the central hub for all users to interact with their sustainability data. It displays real-time information about personal, organizational, and community-level carbon footprints through charts, graphs, and trend indicators. The dashboard supports data filtering by time period, transport mode, and emission intensity, enabling users to identify high-emission activities and potential areas for improvement. Developed using modern web technologies such as HTML and Tailwind CSS, it ensures an interactive and responsive user experience across devices. The dashboard's AI module continuously analyzes the data to predict emission trends, detect anomalies, and suggest optimized travel alternatives. It transforms complex environmental metrics into an intuitive and motivating visualization, bridging the gap between data analytics and user engagement.

5. Promotion of sustainable mobility choices such as electric vehicles, fuel-efficient driving, and shared transport through incentives:
   VayuNetra goes beyond passive monitoring to actively influence positive behavioral change. The platform encourages users to make greener travel decisions by providing context-aware recommendations and real-time suggestions through natural language prompts. For instance, if a user frequently drives short distances alone, the system may suggest nearby carpooling options or public transport routes. Similarly, users who reduce their emissions consistently receive digital recognition and additional credits. Employers can also initiate sustainability challenges or campaigns based on collective employee data. This continuous feedback loop between measurement, recommendation, and reward makes VayuNetra not just a tracking tool but a behavioral transformation platform. It aims to promote low-emission mobility at scale and foster a collaborative ecosystem where individual efforts contribute to broader climate action goals.

In summary, the scope of VayuNetra extends beyond simple carbon emission tracking. It integrates environmental data, AI-driven insights, and a reward-based incentive system into a unified, scalable web application. By combining accurate analytics with user engagement and motivation, the platform seeks to empower individuals and organizations to take measurable steps toward sustainable transportation and a low-carbon future.

Chapter 2

Literature Review

This literature review explores the body of research related to VayuNetra: AI-Powered Carbon Footprint and Emission Tracking System, examining key studies, frameworks, and findings that have shaped the field of AI-based sustainability, emission monitoring, and carbon credit systems. The review provides a critical analysis of existing works, highlighting major contributions in rule-based and data-driven emission estimation, AI for sustainability management, and digital carbon tracking. It also identifies research gaps in real-time tracking, data transparency, and user engagement. By analyzing prior approaches that combine AI, IoT, and behavioral incentive models, this review aims to clarify the current state of sustainable AI research and establish the basis for developing an intelligent, community-driven carbon footprint management system.

2.1 Existing Carbon Emission Estimation Methodologies

The study of carbon emission estimation has evolved considerably over the past two decades, with methodologies shifting from manual inventory-based calculations to automated, AI-assisted frameworks. As climate change and sustainable mobility have become global priorities, researchers have focused on developing techniques that not only quantify emissions but also interpret their causes and implications in real time. This section examines the evolution of emission estimation systems, from rule-based and statistical models to recent AI- and IoT-enabled frameworks, highlighting their methodologies, applications, benefits, and limitations.

2.1.1 From Fuel Ledgers to Digital Carbon Accounting

Historically, emission estimation began with simple fuel-based ledgers, where total fuel consumption was multiplied by emission factors derived from laboratory testing. These methods were foundational but lacked the ability to differentiate emissions across diverse vehicles, routes, or operational conditions. With the rise of computational modeling in the late 20th century, standardized inventory tools emerged to aid transport and energy policy development.

One of the earliest and most influential frameworks was the COPERT (Computer Programme to Calculate Emissions from Road Transport), introduced by the European Envi-

ronment Agency. It provided speed-dependent emission factors, considering parameters such as road gradient, vehicle type, fuel efficiency, and ambient temperature [14]. Similarly, the Greenhouse Gas (GHG) Protocol became the global benchmark for organizational emission reporting, offering three distinct scopes: direct emissions (Scope 1), energy-related emissions (Scope 2), and indirect value-chain emissions (Scope 3) [13].

In India, WRI India, TERI, and CII (2015) localized these methods by developing *India-specific Road Transport Emission Factors*, introducing uplift multipliers to reflect real-world fuel inefficiencies and mixed traffic conditions [12]. Collectively, these advancements transitioned emission measurement from manual estimates to structured, auditable systems that informed both policy and corporate sustainability reporting.

2.1.2 Entry of AI: From Automation to Intelligence

The integration of Artificial Intelligence (AI) marked a major paradigm shift in carbon accounting—from static measurement to predictive and adaptive intelligence. Machine Learning (ML) and Deep Learning (DL) models began to be used to predict emissions, optimize energy usage, and identify high-emission patterns. Roy and Mitra (2020) demonstrated that neural networks could model complex industrial emission datasets for forecasting and reduction planning [1].

Meanwhile, researchers such as Bannour et al. (2021) focused on AI's own carbon footprint, analyzing tools like Carbon Tracker, Experiment Impact Tracker, and Green Algorithms. They found significant discrepancies in reported energy consumption and emissions, calling for standardized digital accounting methods [2]. Budennyy et al. (2022) further extended this field with Eco2AI, an open-source Python package that monitors CO_2 output from ML model training and inference in real time [3]. Bouza (2023) compared several AI carbon estimation tools, revealing that variations in underlying algorithms, hardware efficiency, and data assumptions often produced inconsistent outputs [4]. These developments highlight how AI has not only improved emission prediction accuracy but also brought transparency to digital carbon reporting itself.

2.1.3 Adoption Across Mobility, Homes, and Supply Chains

Beyond industrial applications, emission tracking research expanded into everyday mobility, logistics, and household systems. IoT-based monitoring frameworks became popular for energy efficiency and carbon awareness in homes, integrating sensors and smart meters to provide real-time data visualization [5]. In transport and logistics, AI and IoT systems have been used to optimize routing, manage fleet performance, and reduce operational emissions. Parthasarathy (2024) developed an AI-driven simulation that minimized emissions across logistics networks by optimizing route selection and load distribution [8].

At the individual level, mobile applications began combining GPS tracking with AI-based analytics to calculate personal travel emissions. Jasmy et al. (2024) demonstrated how behavioral nudges, delivered through a mobile app, could improve users' awareness of their carbon impact and influence sustainable travel decisions [9]. Despite such promise, most implementations remained small-scale prototypes or simulations, highlighting challenges in data reliability, scalability, and real-world adoption [6][11].

2.1.4 Benefits and Opportunities

The reviewed literature reveals several consistent benefits of integrating AI and IoT into emission estimation systems: (i) Enhanced Granularity: AI-based systems process diverse datasets—distance, mode, occupancy, route condition, and temporal context—to generate precise, situation-aware emission outputs [1][8][11]. (ii) Real-Time Responsiveness: IoT-enabled devices and mobile applications allow continuous emission monitoring, facilitating immediate behavioral feedback and adaptive policy mechanisms [5][9]. (iii) Personalization and Communication: Natural Language Processing (NLP) and adaptive interfaces convert technical CO_2 metrics into user-friendly, actionable insights that encourage public participation [9]. (iv) Data-Driven Governance: By aligning AI-based emission tracking with frameworks such as the GHG Protocol, governments and corporations can maintain data traceability, accountability, and compliance with international sustainability goals [12][13]. (v) Collective Optimization: Multi-agent simulations in logistics and urban mobility reveal that emission reductions scale significantly when collective behavior and network-level optimizations are applied concurrently [11][15].

2.1.5 Challenges, Ethics, and Practical Limits

Despite technological advances, several challenges persist across existing methodologies. One major concern is the variance among emission calculators, which often produce divergent results due to inconsistent assumptions and incomplete lifecycle coverage [2][4][10]. The data dependency of AI systems requires high-resolution telemetry, which raises cost, privacy, and interoperability issues [1][8].

Moreover, many emission factors remain geographically biased. Models like COPERT are calibrated primarily for European traffic and fuel standards, limiting their direct applicability in developing nations [12][14]. Another recurring issue is behavioral reliability—users may unintentionally or intentionally misreport data in incentive-based systems [9][13]. Lastly, most frameworks have been validated only in controlled environments or pilot studies, indicating a significant deployment gap between research outcomes and city-scale implementation [8][11][15].

2.1.6 Existing Methodologies for Carbon Emission Estimation

Multiple standardized methodologies exist for calculating carbon emissions across industries and transport systems. These include:

- GHG Protocol (Scope 3): Employs fuel, distance, and spend-based estimation techniques for business-related emissions, emphasizing organizational reporting [13].
- COPERT II: Calculates emissions based on speed, road gradient, vehicle type, and ambient temperature; extensively used in European environmental monitoring [14].
- India-Specific Road Transport Factors (WRI/TERI/CII): Provides regionally calibrated emission factors for passenger and freight vehicles under Indian conditions [12].

While these frameworks offer robust accuracy for industrial and governmental reporting, they are complex and data-intensive, often requiring detailed engine performance, temperature, and fuel-composition data. As a result, they are less suited for real-time, user-level

tracking or community-based carbon management. Simplified distance-based models, although easier to apply, fail to incorporate dynamic elements such as vehicle age, occupancy, air quality, and localized weather patterns.

2.1.7 Direction of Research and Emerging Trends

Recent research trends reveal a decisive movement toward simplified yet context-aware carbon accounting models. Studies focus on integrating real-time mobility data, IoT sensors, and AI analytics to improve accessibility and engagement. Mobile-based platforms are being developed to estimate trip-level emissions, gamify sustainable behavior, and visualize community-scale impact [8][9].

Advances in cloud computing, edge analytics, and open data initiatives are enabling more efficient emission tracking architectures that balance precision with computational feasibility. Furthermore, growing interest in community-level dashboards and credit-based sustainability programs demonstrates how digital participation can enhance collective accountability. Researchers increasingly advocate for localized models that account for regional driving behaviors, fuel quality, and urban conditions, thereby improving contextual validity.

2.1.8 Discussion and Research Gaps

The literature highlights several unaddressed areas that require further investigation. Most existing emission estimation systems are oriented toward industrial or organizational reporting, with minimal focus on individual or transport-level analysis. Studies involving AI and ML emphasize predictive performance but often overlook model interpretability and user comprehension. Another critical gap lies in the limited integration of behavioral incentives—such as credit mechanisms or feedback systems—into emission management frameworks, even though multiple studies confirm their effectiveness in promoting sustainable actions [9][11]. Finally, there is a notable scarcity of real-world, large-scale validations of AI-based emission tracking platforms. The lack of standardized datasets, cross-regional benchmarks, and open-access methodologies constrains reproducibility and scalability. Addressing these research gaps could advance the development of transparent, adaptive, and participatory carbon tracking ecosystems that link individual behavior with global sustainability outcomes.

[**1**] Soumit Roy and Mainak Mitra presented AI Driven Strategies for Carbon Footprint Reduction — A Scholarly Exploration, which proposed an AI-based framework using machine learning and recurrent neural networks to analyze industrial emission data and suggest reduction strategies. While the model demonstrated the potential of AI in sustainability forecasting, it required extensive data, computational resources, and had limited real-world deployment feasibility.

[**2**] Nesrine Bannour et al., in their work Evaluating the Carbon Footprint of NLP Methods: A Survey and Analysis of Existing Tools, compared several emission measurement tools such as Carbon Tracker, Experiment Impact Tracker, and Green Algorithms. Their study emphasized the importance of transparent carbon reporting in AI experiments but found that many tools underestimate real-world emissions and lack precision for small-scale experiments.

[**3**] S. A. Budennyy et al. proposed Eco2AI: Carbon Emissions Tracking of Machine Learning Models, an open-source Python tool that monitors the CO_2 output of AI model training and inference. The tool made AI carbon tracking more accessible but primarily focused on computational emissions, neglecting broader lifecycle or transport-based emissions.

[**4**] Lucía Bouza et al. developed a detailed review titled How to Estimate Carbon Footprint When Training Deep Learning Models? A Guide and Review, comparing different carbon estimation frameworks. The study highlighted the inconsistency between emission calculators and called for unified standards and datasets to improve transparency and reproducibility.

[**5**] Lanre Olatomiwa et al. presented A Review of IoT-Based Visualization Platforms for Tracking Household Carbon Footprints, analyzing smart home and IoT integration for carbon monitoring. The paper showed that IoT dashboards can encourage energy-efficient habits, but most models were limited to household contexts, lacking large-scale, real-time community engagement.

[**6**] Aurora Martiny in her master's thesis Towards Sustainable AI: Monitoring and Analysis of Carbon Emissions in Machine Learning Algorithms, used empirical data from LSTM models to measure training emissions through the CodeCarbon framework. Although the study provided in-depth energy consumption metrics, it was restricted to a single environment and lacked scalability across domains.

[**7**] Uzair Aslam Bhatti et al. in their paper Artificial Intelligence Applications in Reduction of Carbon Emissions: Step Towards Sustainable Environment, provided an overview of AI's role in emission reduction across industrial and urban domains. While insightful, the paper lacked experimental validation and served primarily as a conceptual discussion of sustainable AI.

[**8**] VenkateshPrabu Parthasarathy published AI-Driven Carbon Footprint Tracking and Emission Reduction in Logistics Networks, which introduced an AI- and IoT-integrated model to optimize logistics routes for emission reduction. The system showed potential in improving route efficiency and reducing fuel consumption, but remained a simulation-based

model with no real-world deployment.

[**9**] Ahmad Jasim Jasmy et al. developed a mobile-based system titled A Novel Approach to Sustainable Behavior Enhancement Using AI-Driven Carbon Footprint Assessment and Real-Time Analytics. The study demonstrated that AI-driven behavioral nudges can promote eco-friendly practices among users, but the results were limited by a small dataset and short observation period.

[**10**] Mina Vildan Şark in her research Carbon Footprint of Artificial Intelligence Models: Estimation and Reduction Approaches, reviewed carbon estimation and reduction strategies used in AI model training. Through interviews and case studies, she found that most organizations lacked standardized reporting mechanisms and complete visibility of their carbon data.

[**11**] Rong Huang and Shuai Mao introduced Carbon Footprint Management in Global Supply Chains: A Data-Driven Approach Utilizing Artificial Intelligence, proposing an AI and CNN-based framework for emission prediction across logistics networks. The model achieved accurate predictions but required extensive data and computational resources, making real-time implementation challenging.

[**12**] WRI India, TERI, and CII developed India-Specific Road Transport Emission Factors, formulating an emission estimation model using the equation: Emission Factor (kg CO_2/km) = (Fuel Emission Factor / Fuel Efficiency). The model also included an uplift factor (8–10%) for real-world conditions but lacked regional customization and real-time adaptability.

[**13**] The Greenhouse Gas Protocol introduced the Technical Guidance for Calculating Scope 3 Emissions (Category 6: Business Travel), providing fuel-based, distance-based, and spend-based methods for calculating transport-related CO_2 emissions. However, the model was mainly designed for corporations and not for individual tracking or community-scale applications.

[**14**] The European Environment Agency proposed the COPERT II Methodology for Road Transport Emission Calculations, a comprehensive model incorporating parameters like vehicle speed, road gradient, and vehicle type. Although highly detailed, the model's complexity and European-specific parameters make it difficult to directly apply to Indian transport conditions.

[**15**] The recent report AI-Powered Carbon Tracking in Smart Cities highlighted the use of AI-integrated platforms for real-time monitoring of urban transport emissions and sustainable mobility. While this research introduced a promising vision for future smart cities, it focused primarily on infrastructure-level deployment rather than individual behavioral participation.

2.2 Comparative Analysis of Literature Survey

Table 2.1: Comparative Analysis of Literature Survey

Sr. No.	Title	Author(s)	Year	Methodology	Drawback
1	AI Driven Strategies For Carbon Footprint Reduction — A Scholarly Exploration	Soumit Roy, Mainak Mitra	2020	AI-driven framework using ML and RNNs with IoT/sensor data for emission reduction.	High data requirements, high computation cost, and deployment challenges.
2	Evaluating the Carbon Footprint of NLP Methods: A Survey and Analysis of Existing Tools	Nesrine Bannour, Sahar Ghannay, Aurélie Névéol, Anne-Laure Ligozat	2021	Survey and evaluation of six tools (Carbon Tracker, Experiment Impact Tracker, Green Algorithms, ML CO_2 Impact, Energy Usage, Cumulator) on NER experiments.	Tools underestimate emissions, ignore hardware lifecycle, vary greatly, and lack precision for small experiments.
3	AI Carbon Emissions Tracking of Machine Learning Models as the First Step Towards Sustainable AI	S. A. Budennyy	2022	Introduces open-source Python tool eco2AI, tested on large models (DALL·E, GPT, etc.).	Tracks only training/inference, ignores embodied or full lifecycle emissions.
4	How to Estimate Carbon Footprint When Training Deep Learning Models? A Guide and Review	Lucía Bouza	2023	Comparative review and experimental evaluation of AI carbon tracking tools.	Tool variance, focus mainly on training; lacks process specificity.
5	A Review of IoT Based Visualization Platforms for Tracking Household Carbon Footprints	Lanre Olatomiwa	2023	Literature review of IoT visualization platforms and scenario analysis.	Focused mainly on household platforms; lacks empirical comparison.
6	Towards Sustainable AI: Monitoring and Analysis of Carbon Emissions in Machine Learning Algorithms	Aurora Martiny	2023	Empirical experiments and parameter studies on LSTMs; evaluation using CodeCarbon.	Limited to time-series data, single location; tool limitations.

Sr. No.	Title	Author(s)	Year	Methodology	Drawback
7	Artificial Intelligence Applications in Reduction of Carbon Emissions: Step Towards Sustainable Environment	Bhatti	2024	Editorial overview of recent AI approaches in emission reduction.	Lacks primary data or empirical evaluation; descriptive summary only.
8	AI-Driven Carbon Footprint Tracking and Emission Reduction in Logistics Networks	VenkateshPrabu Parthasarathy	2024	Simulation-based study using AI/ML/IoT to optimize logistics emissions.	Simulation only, not real deployment; high setup cost; data sensitivity.
9	A Novel Approach to Sustainable Behavior Enhancement through AI-Driven Carbon Footprint Assessment and Real-Time Analytics	Ahmad Jasim Jasmy et al.	2024	Empirical mobile app study for real-time tracking and behavioral intervention (TRA model).	Small user base; limited statistical significance for attitude change.
10	Carbon Footprint of Artificial Intelligence (AI) Models: Estimation and Reduction Approaches	Mina Vildan	2025	Literature review, case studies, and expert interviews on AI carbon estimation.	Lack of standardization and incomplete data; inconsistent measurement.
11	Carbon Footprint Management in Global Supply Chains: A Data-Driven Approach Utilizing Artificial Intelligence	Rong Huang, Shuai Mao	2025	Data-driven framework using CNN and optimization for emission prediction and supply chain analysis.	Heavy data requirements, high computation cost, limited interpretability.
12	India Specific Road Transport Emission Factors	WRI India, TERI, CII	2015	Developed India-specific emission factors for passenger, freight, and bus transport using the formula: Emission Factor (kg CO_2/km) = (Fuel Emission Factor / Fuel Efficiency), with an uplift factor (8–10%) for real-world driving.	Lacks region-specific data and real-time accuracy; assumes constant fuel efficiency.

Sr. No.	Title	Author(s)	Year	Methodology	Drawback
13	Technical Guidance for Calculating Scope 3 Emissions (Category 6: Business Travel)	WRI, GHG Protocol	2013	Provides three methods — Fuel-based, Distance-based, and Spend-based — for calculating transport-related CO_2 emissions; Distance method uses CO_2e = (distance × emission factor) across travel modes.	Corporate-focused; not designed for individual or community emission tracking.
14	COPERT II: Road Transport Emission Calculation Methodology	EEA, Ntziachristos & Samaras	1997	Establishes COPERT model for vehicle emission estimates using speed-dependent factors, road gradient, weight, and driving condition parameters.	Complex setup; European driving context limits direct Indian application.

Chapter 3

Project Design

The project design of **VayuNetra** focuses on building an integrated, AI-enabled platform for transportation-based carbon footprint management. It combines transparent, rule-based emission models with geolocation context and natural-language insights to deliver accurate trip-level calculations, real-time sustainability tips, and community progress tracking. A clean web interface with role-based access (employees, employers, banks, admins) supports secure data entry, dashboards, and a digital carbon-credit wallet to motivate greener choices. The design emphasizes accessibility, scalability, and responsive performance so the system can be effectively used across homes, campuses, workplaces, fleets, and city programs.

3.1 Existing System

Most existing carbon footprint tracking systems focus mainly on estimating emissions at a broad or organizational level, often overlooking the individual and transportation-based perspective that directly reflects daily lifestyle choices. Many current tools calculate emissions using generic averages or region-wide data, without accounting for crucial personal factors such as travel mode, fuel type, vehicle age, occupancy, and distance traveled. As a result, users receive approximate or generalized estimates rather than accurate, personalized insights into their real environmental impact.

Another major limitation of existing systems is their dependence on manual data entry or complex IoT setups, making them less accessible to the general public. Many solutions also lack real-time awareness features, such as geolocation-based pollution context or dynamic adjustments for route conditions and industrial proximity. Additionally, current carbon-tracking applications seldom integrate AI or natural language processing (NLP) to provide actionable suggestions or personalized eco-friendly recommendations.

Furthermore, most systems do not include a reward-based mechanism, such as digital carbon credits, to motivate users to adopt sustainable travel habits. The absence of community dashboards and comparative analytics also restricts collaborative efforts toward collective emission reduction. Overall, existing systems remain fragmented, data-heavy, and non-interactive—highlighting the need for a more intelligent, transparent, and user-centered platform like VayuNetra, which bridges the gap between accurate emission tracking, actionable insights, and behavioral motivation.

3.2 Proposed System

The proposed system architecture for **VayuNetra** is an AI-powered web-based platform designed to monitor, analyze, and reduce transportation-related carbon emissions. It integrates multiple modules — Carbon Footprint Calculation, AI Sustainability Insights, Carbon Credit and Wallet System, and Community Dashboards — to deliver a complete and interactive sustainability management solution.

The Carbon Footprint Calculation module uses a transparent, rule-based model to estimate emissions based on parameters such as travel mode, distance, fuel type, vehicle age, occupancy, and proximity to industrial zones. The AI Sustainability Insights module employs natural language processing (NLP) to provide personalized eco-friendly tips and real-time guidance to users, helping them make greener travel decisions. The Carbon Credit and Wallet System rewards users—especially employees and employers—for adopting sustainable practices by converting their emission reductions into redeemable digital credits. The Community Dashboards module aggregates user and organizational data to display group progress, leaderboards, and comparative analytics that promote collective environmental action.

Together, these modules form an intelligent, data-driven ecosystem that transforms carbon tracking into an engaging and motivational experience. **VayuNetra** not only helps users understand their individual environmental impact but also drives long-term behavioral change through awareness, incentives, and community participation—contributing toward India's broader goals of achieving carbon neutrality and sustainable urban living.

3.2.1 Critical Components of System Architecture

The proposed system architecture for **VayuNetra** integrates multiple modules — Carbon Footprint Calculation, AI Sustainability Insights, Carbon Credit and Wallet System, and Community Dashboards — to simplify and automate transportation-based carbon emission tracking. The system collects user inputs such as travel mode, fuel type, distance, and vehicle condition, then uses rule-based algorithms to calculate emissions accurately. The AI Sustainability Insights module provides real-time, personalized recommendations using natural language processing, helping users adopt greener choices. The Carbon Credit and Wallet System converts eco-friendly actions into digital rewards, motivating both employees and employers to participate in emission reduction. Finally, the Community Dashboards module visualizes collective progress through leaderboards, reports, and analytics that encourage collaboration. Together, these components create an intelligent, scalable, and user-friendly platform that promotes sustainable transportation habits and contributes to long-term environmental awareness.

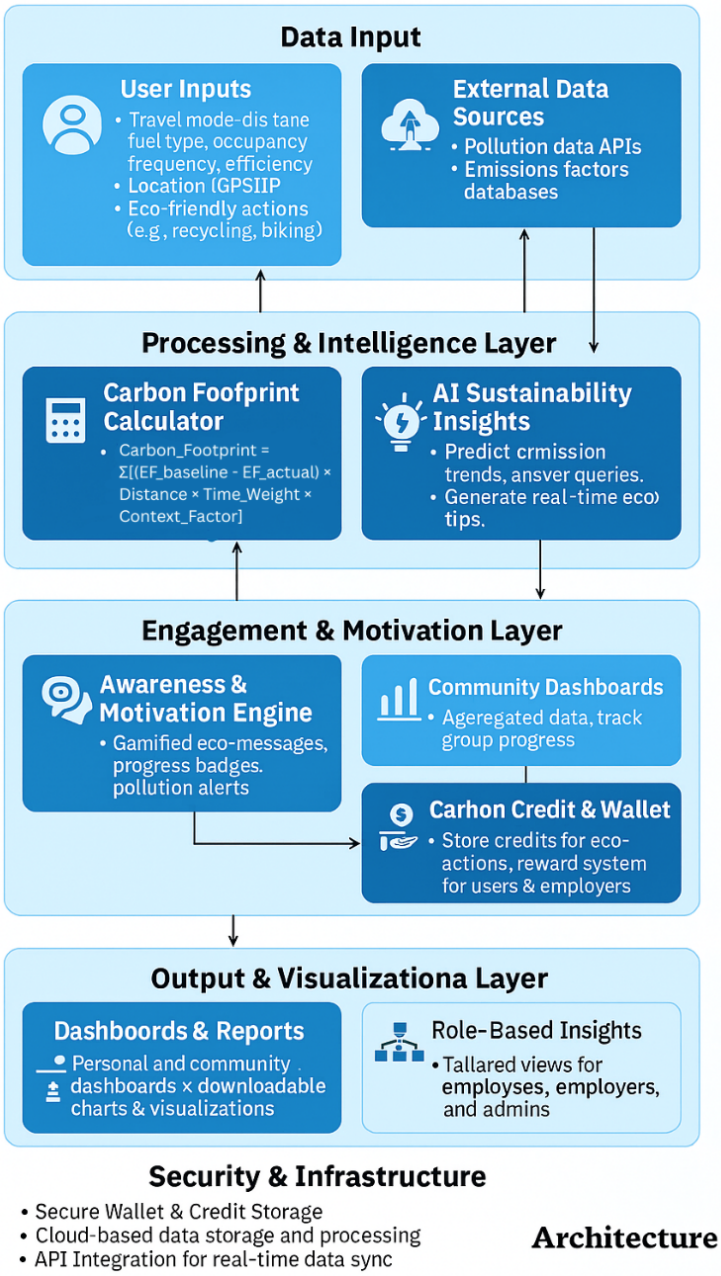

Figure 3.1: System Architecture of VayuNetra

The overall architecture of the proposed system is shown in Figure 3.1. It represents how different modules of VayuNetra interact to collect user and external data, calculate emissions, and provide intelligent insights. The system processes user inputs through the Carbon Footprint Calculator and AI Sustainability Insights, then engages users through dashboards, rewards, and motivation tools. Finally, the results are visualized through reports and dashboards while maintaining secure data storage and processing.

3.2.2 Carbon Emission and Credit Calculation

This section explains the logic and step-by-step computation used by *VayuNetra* to calculate carbon credits (CC) earned by users. The calculation takes into account the user's travel mode, emission factors, distance covered, and contextual conditions such as weather, air quality, and traffic. These parameters help the system generate fair and realistic carbon credit values for each trip.

Parameters Used:

- **CC:** Carbon Credits earned by the user.

- **$\mathbf{EF_{baseline}}$:** Baseline emission factor representing the standard emission rate per km (e.g., petrol car).

- **$\mathbf{EF_{actual}}$:** Actual emission factor based on the user's chosen commute mode (e.g., electric bike, carpool, hybrid car).

- **Distance:** The distance traveled during the commute.

- **Time_Weight:** A weighting factor accounting for time-related factors such as peak vs. off-peak hours, trip frequency, and recency.

- **Context_Factor:** Adjustment based on contextual parameters such as weather, traffic load, nearby industrial zones, and AQI levels.

Step-by-Step Calculation Example:

Step 1: Emission Factors
$EF_{baseline}$: 0.12 kg CO_2/km (cycling baseline)
EF_{actual}: 0.00 kg CO_2/km (zero emissions)
Savings: 0.12 kg CO_2/km

Step 2: Distance
Distance: 17.55 km
Duration: 54.7 minutes
Average Speed: 19.2 km/h

Step 3: Time Weight
Time_Weight = Peak_Factor × Traffic_Multiplier × Recency_Weight
= 1.2 × 1.0 × 1.0 = 1.2
(Representing peak evening, light traffic, and recent trip conditions)

Step 4: Context Factor
Context_Factor = Weather × Route × Load × AQI × Seasonal

$= 0.95 \times 1.2 \times 1.0 \times 1.1 \times 0.95 = 1.254$
(Favorable weather, city center route, poor AQI, winter season)

Step 5: Final Calculation

$$CC = (EF_{baseline} - EF_{actual}) \times Distance \times Time_Weight \times Context_Factor$$
$$CC = (0.12 - 0.00) \times 17.55 \times 1.2 \times 1.254$$
$$CC = 3.1691 \text{ kg } CO_2$$

Result: The cycling trip earned approximately 3.17 kg CO_2 credits — equivalent to avoiding nearly 24.4 km of petrol car travel.

This rule-based model ensures that each user's trip is evaluated dynamically, rewarding low-emission behaviors such as walking, cycling, or carpooling with proportionally higher carbon credits. By incorporating contextual factors and real-time data, the calculation provides both accuracy and fairness in assessing an individual's environmental impact.

3.3 System Diagrams

The System Diagrams of VayuNetra: AI-Powered Carbon Footprint and Emission Tracking System illustrate the complete interaction between various functional modules and data components that together enable intelligent emission estimation, analysis, and sustainability management. These diagrams act as a visual representation of the system's logical architecture, showing how user input and contextual data flow through multiple interconnected processes to generate meaningful environmental insights.

The system begins with the collection of essential input data, including user trip details such as travel mode, fuel type, distance, vehicle condition, and proximity to industrial zones. This data, entered manually by users or automatically derived from geolocation APIs, passes through the Preprocessing and Validation Layer, which ensures input accuracy, normalization, and consistency. The validated data is then processed by the Rule-Based Emission Calculator, which computes the CO_2 output per trip using predefined emission factors and context-aware adjustment parameters.

The computed results are further analyzed in the Emission Intelligence Layer, where AI-driven algorithms interpret the emission data to generate insights, sustainability scores, and personalized suggestions for greener alternatives such as cycling, carpooling, or public transport. These outputs are visualized through the User Dashboard and Reporting Module, which provides clear, interactive charts displaying emission trends, credit balances, and community-level progress.

Additionally, the system integrates a Carbon Credit and Wallet System that rewards users for eco-friendly behavior by converting emission savings into tradable credits. The Admin and Employer Portals facilitate organizational oversight, allowing monitoring of aggregated emissions and sustainability performance across departments or communities. The Authentication and Role Management Layer ensures that users—employees, employers, banks, and administrators—securely access only the modules relevant to their roles.

Overall, the System Diagrams of VayuNetra encapsulate how its interconnected modules work cohesively to convert raw mobility and environmental data into actionable

insights, rewards, and awareness tools. This architecture highlights the platform's ability to promote behavioral change and support a collaborative approach toward reducing transportation-related carbon emissions.

3.3.1 UML Diagram

Unified Modeling Language (UML) is a general-purpose modeling language used to define a standard way to visualize the structure and design of a system. UML is closely associated with object-oriented design and analysis. It employs various elements and relationships to form diagrams that represent the system's structure and behavior.

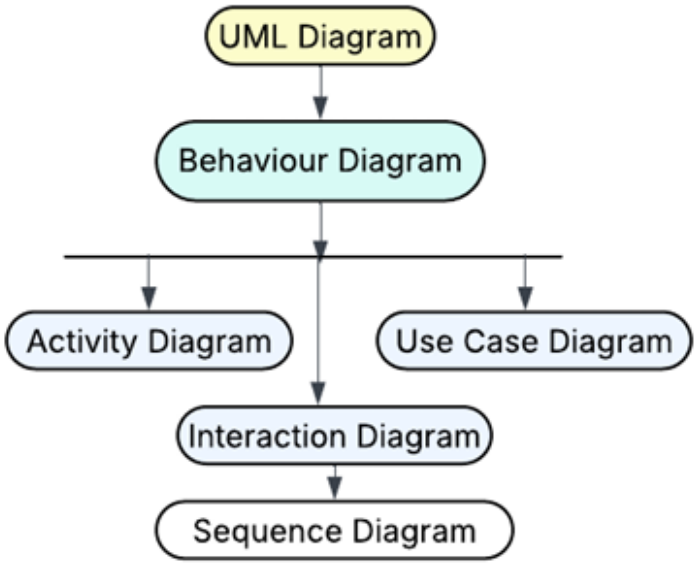

Figure 3.2: UML Diagram of Vayunetra System

As shown in Figure 3.2, the UML diagram visualizes the interaction between the system's components and their roles in achieving seamless communication and function execution.

3.3.2 Activity Diagram

The activity diagram of the VayuNetra system illustrates the overall workflow of carbon emission tracking, from user login to emission calculation and credit generation. The process begins when the user logs into the platform and records a trip by entering transport details such as mode, fuel type, and distance. The system then retrieves the relevant emission factors, computes the carbon footprint using rule-based formulas, and stores the result in the database. Based on the calculated emissions, carbon credits are awarded and reflected in the user's dashboard. The flow concludes with data visualization and report generation for further analysis. Activity diagrams illustrate the flow of control in a system. They are used to represent the sequence of actions or activities involved in executing a use case.

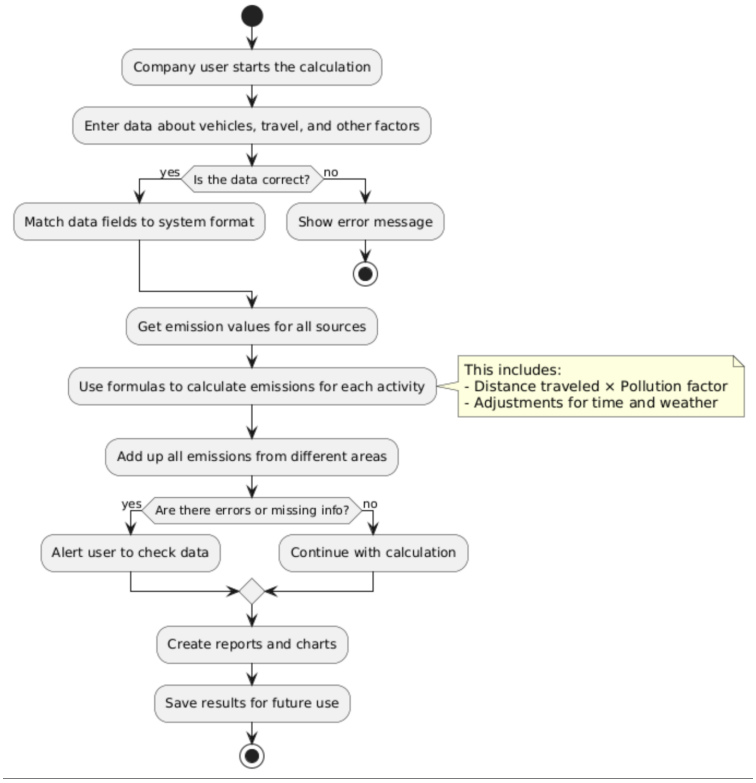

Figure 3.3: Activity Diagram for carbon footprint calculation system

The activity diagram illustrates the workflow for the Carbon Footprint Calculation System in the VayuNetra platform. The process begins when a user—typically a company representative or an individual—initiates the emission calculation. The user first enters necessary information such as travel details, vehicle type, fuel type, distance covered, and other relevant environmental factors. The system then validates the entered data; if the data is found to be incorrect or incomplete, the user is notified to correct the input before proceeding. Once the data is verified, it is matched to the system's required format, and emission values are fetched from predefined databases or emission factor tables.

The system then uses a rule-based formula to calculate emissions for each activity, primarily using the equation: Emission = Distance Traveled × Pollution Factor, with possible adjustments for factors like time of travel, vehicle efficiency, and nearby industrial pollution levels. After calculating emissions for all sources, the system aggregates the results to obtain the total carbon footprint. If inconsistencies or missing information are detected, the user is alerted to review and correct the data. Once validated, the system continues to generate reports and charts that summarize emission patterns, helping users visualize their environmental impact. Finally, the results are saved for future reference and can be used for trend analysis, comparison, or generating carbon credit insights. This structured workflow ensures accuracy, transparency, and repeatability in emission tracking and reporting.

3.3.3 Use Case Diagram

A Use Case Diagram is a visual representation of how users interact with a system. It identifies the various types of users (actors) and the specific functions or services they can perform. This diagram helps in understanding the system's overall functionality and defining user requirements clearly. In the case of VayuNetra, the use case diagram illustrates how different users such as employees, employers, banks, and administrators interact with the platform for activities like trip logging, emission tracking, and credit management.

The Use Case Diagram in Figure 3.4 illustrates the **VayuNetra: AI-Powered Carbon Footprint and Emission Intelligence System**, highlighting its main functionalities and the interactions between different user roles and system components. The system is designed around three primary user categories—*Individual User*, *Employer User*, and *Carbon Offset Admin*—each performing distinct operations within the carbon credit management framework. Individual users can register, submit travel and emission data, and view personal credits or reports. Employers can generate and trade carbon credits based on organizational data, while the admin oversees credit approvals, user management, and system integrity. This diagram defines the boundaries of the VayuNetra system and showcases how various roles collaborate to ensure accurate emission tracking, transparent credit management, and incentive-driven sustainability participation.

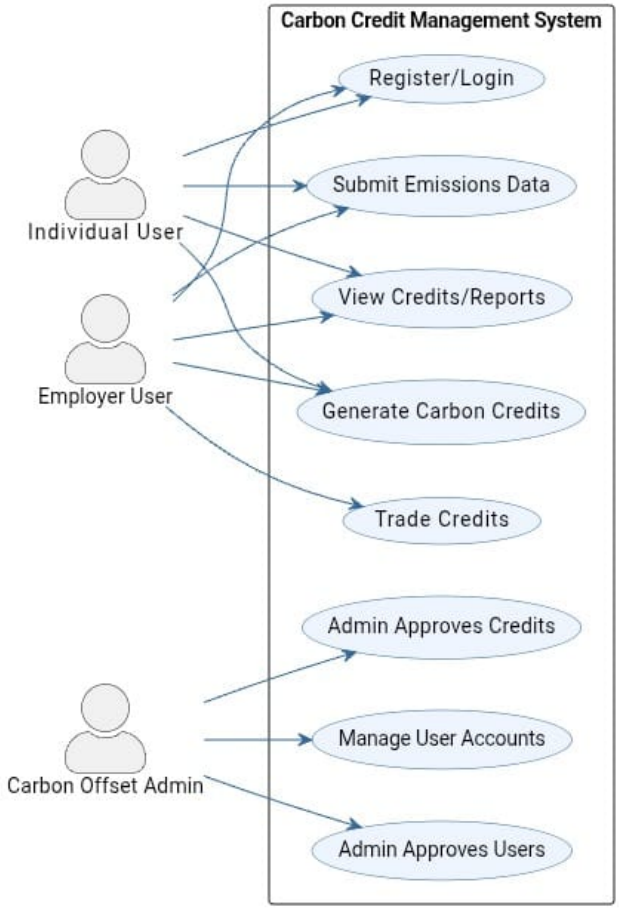

Figure 3.4: Use Case Diagram of VayuNetra System

The diagram represents the main interactions between different users and the VayuNetra system. It shows how Individual Users, Employer Users, and Carbon Offset Admins interact with various modules of the carbon credit management platform.

- **Individual User:** Can register or log in to the system, submit travel and emission data, and view personal credits and reports. This allows users to monitor their carbon footprint and understand their environmental impact.
- **Employer User:** Has access to generate carbon credits based on collective employee data, view reports, and trade credits within the system to balance emissions. Employers play a key role in promoting sustainable practices among employees.
- **Carbon Offset Admin:** Responsible for verifying and approving carbon credits generated by users or organizations, managing user accounts, and overseeing system-level activities to maintain data integrity and transparency.

Together, these interactions ensure a transparent, efficient, and incentive-based carbon credit ecosystem where users can track, trade, and reduce emissions effectively.

3.3.4 Sequence Diagram

A Sequence Diagram shows how objects or components in a system communicate with each other over time. It represents the sequence of messages exchanged between the entities to complete a particular process or workflow. In VayuNetra, the sequence diagram demonstrates how user inputs, emission calculations, and credit updates flow through the system, ensuring smooth data processing and accurate carbon footprint computation. The sequence diagram in Figure 3.5 illustrates the detailed interaction flow between different actors and modules within the VayuNetra system during the carbon emission tracking and credit management process.

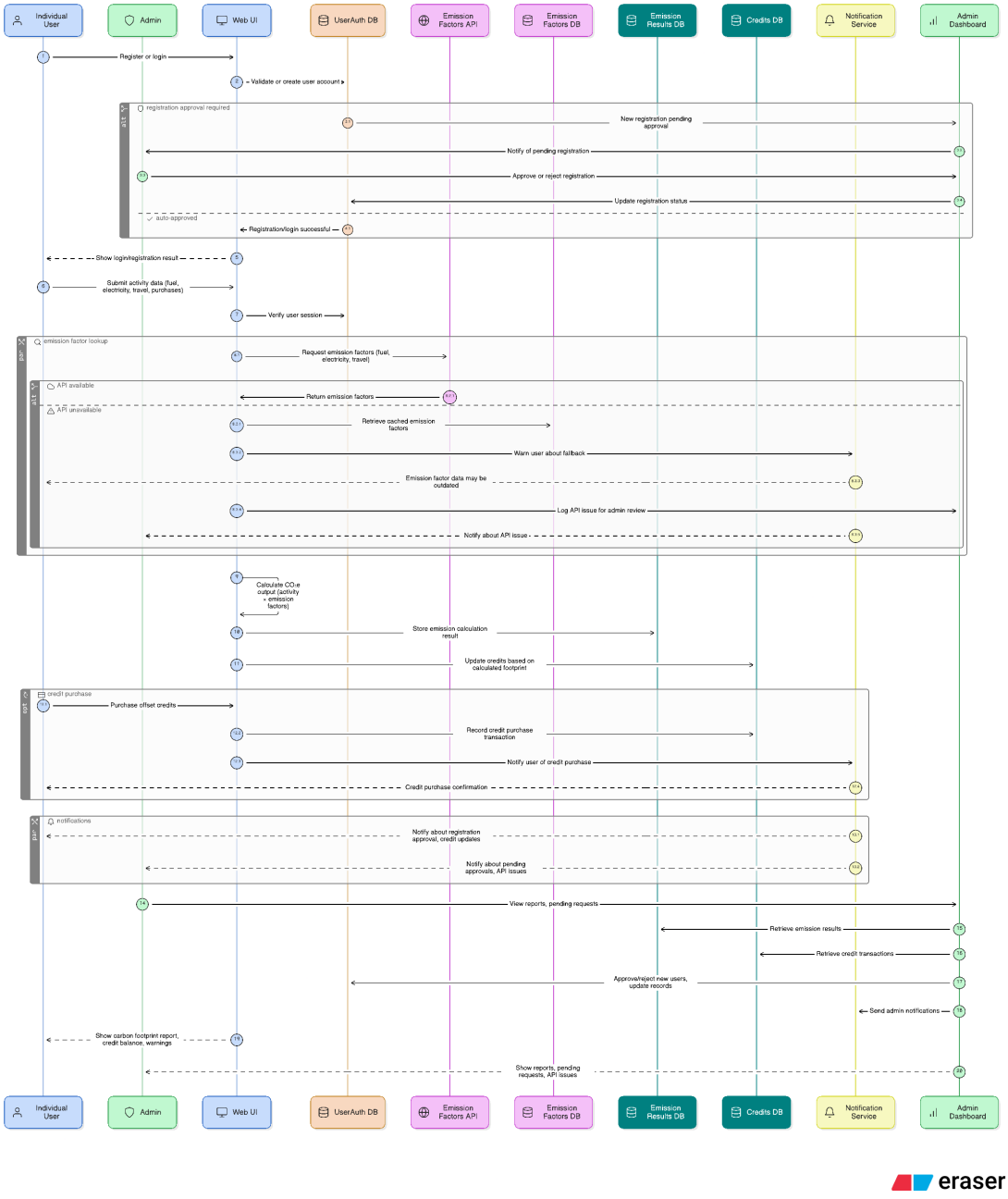

Figure 3.5: Sequence Diagram of VayuNetra System

The process begins with the Individual User registering or logging into the system through the Web Interface. The registration details are validated by the Admin and stored in the User Authentication Database. Once verified, the user submits travel and emission-related data such as vehicle type, fuel used, and distance traveled. The system retrieves emission factors from the Emission Factors Database or through the Emission Factors API and computes total emissions using rule-based formulas. The calculated results are then stored in the Emission Results Database.

Based on the computed emissions, the system updates the user's available credits in the Credits Database. Users or employers can then purchase or trade credits through the platform, and all such transactions are logged for transparency. The Admin Dashboard continuously monitors these activities, approving new registrations, emission records, and credit transactions. Finally, the system sends notifications about approvals, reports, or pending actions through the Notification Service. The overall workflow ensures secure data handling, transparency in carbon credit management, and seamless coordination between users, admins, and system modules.

Chapter 4

Project Implementation

This chapter presents the core aspects of the implementation, including essential feature-specific code snippets and the overall project timeline. It illustrates how the system modules were developed, how different components were integrated, and how the AI models were applied to real-time data processing. The timeline highlights the structured progression of development, while the code snippets emphasize technically critical parts of the implementation.

4.1 Code Snippets

This section showcases the important parts of the implementation for the *VayuNetra: AI-Powered Carbon Footprint and Emission Intelligence System*. It highlights how the models for storing emission factors, logging user trips, and calculating emission records were structured and integrated within the Django framework. Each snippet represents a core functionality essential to the operation of the platform, enabling accurate emission tracking, AI-driven analysis, and data management within the system.

The following code snippets provide an overview of the backend logic that drives the main functionalities of VayuNetra, including emission data storage, rule-based emission computation, and initialization of baseline factors. Each snippet contributes to the seamless workflow of data collection, processing, and carbon footprint computation.

```python
# app/models.py
from django.db import models
from django.contrib.auth.models import User

class EmissionFactor(models.Model):
    """
    Stores gCO2 per km for a (mode, fuel) pair.
    Example rows:
      ("car", "petrol") => 192.0
      ("car", "diesel") => 171.0
      ("bus", "diesel") => 82.0
      ("train", "electric") => 25.0  # grid-mix dependent
    """
    MODE_CHOICES = [
        ("car", "Car"), ("carpool", "Carpool"), ("bus", "Bus"),
        ("train", "Train"), ("bike", "Bike"), ("walk", "Walk")
    ]
    FUEL_CHOICES = [
        ("petrol", "Petrol"), ("diesel", "Diesel"),
        ("electric", "Electric"), ("na", "Not Applicable")
    ]
    mode = models.CharField(max_length=16, choices=MODE_CHOICES)
    fuel = models.CharField(max_length=16, choices=FUEL_CHOICES, default="na")
    factor_g_per_km = models.FloatField(help_text="grams CO2 per km per passenger")

    class Meta:
        unique_together = ("mode", "fuel")

    def __str__(self):
        return f"{self.mode}/{self.fuel}: {self.factor_g_per_km} g/km"

class Trip(models.Model):
    """
    One commute or travel segment logged by a user.
    """
    user = models.ForeignKey(User, on_delete=models.CASCADE)
    date = models.DateField()
    mode = models.CharField(max_length=16)
    fuel = models.CharField(max_length=16, default="na")
    distance_km = models.FloatField()
    occupancy = models.PositiveIntegerField(default=1)  # carpool headcount etc.
    vehicle_age_years = models.PositiveIntegerField(default=0)
    near_industrial_zone = models.BooleanField(default=False)

class EmissionRecord(models.Model):
    """
    Stores calculated emissions in kgCO2 for a Trip snapshot.
    """
    trip = models.OneToOneField(Trip, on_delete=models.CASCADE)
    kg_co2 = models.FloatField()
    notes = models.TextField(blank=True, default="")
    created_at = models.DateTimeField(auto_now_add=True)
```

Figure 4.1: Django Models for Emission Factors, Trip Data, and Emission Records

Before performing any emission calculations, VayuNetra defines its foundational data structure through Django models. These models represent the building blocks of the database schema that power all emission-related operations.

This section defines VayuNetra's core Django models for emission tracking. The EmissionFactor model stores baseline emission values (gCO_2/km per person) for each transport mode and fuel type. The Trip model records user travel details such as distance, mode, fuel, and vehicle condition, serving as input for emission calculations. The EmissionRecord model stores the computed results in $kgCO_2$ for each trip, enabling progress tracking and report generation. Together, these models support accurate emission computation and structured data storage.

In summary, these models form the foundation of the system's database layer, allowing VayuNetra to efficiently manage user trips, emission factors, and computed results for both analytics and reporting purposes.

```python
# app/services/emission_calc.py
from dataclasses import dataclass

@dataclass
class TripInput:
    mode: str
    fuel: str
    distance_km: float
    occupancy: int = 1
    vehicle_age_years: int = 0
    near_industrial_zone: bool = False
    base_factor_g_per_km: float = 0.0  # looked up from DB

def age_penalty(age_years: int) -> float:
    """
    Older vehicles typically emit more due to wear/inefficiency.
    +1% per year after year 3, capped at +15%.
    """
    extra = max(age_years - 3, 0) * 0.01
    return min(extra, 0.15)

def industrial_multiplier(near_zone: bool) -> float:
    """
    Awareness context: if route passes near an industrial zone (esp. night),
    we show a +5% context multiplier (not 'real' tailpipe increase; used for awareness).
    """
    return 1.05 if near_zone else 1.0

def per_person_divisor(mode: str, occupancy: int) -> int:
    """
    Share emissions per passenger for carpool, taxi-share etc.
    """
    if mode in ("carpool",) and occupancy > 0:
        return occupancy
    return 1

def compute_emission_kg(trip: TripInput) -> float:
    """
    Core formula (per person):
      kgCO2 = (distance_km * base_factor_g_per_km / 1000)
              * (1 + age_penalty) * industrial_multiplier
              / per_person_divisor
    """
    base_kg = trip.distance_km * (trip.base_factor_g_per_km / 1000.0)
    adj = (1 + age_penalty(trip.vehicle_age_years)) * industrial_multiplier(trip.near_industrial_zone)
    share = per_person_divisor(trip.mode, trip.occupancy)
    return round((base_kg * adj) / max(share, 1), 4)
```

Figure 4.2: Rule-based Emission Calculator and Adjustment Functions

After defining the data models, the next essential step is to calculate carbon emissions accurately based on the user's trip data. The following snippet illustrates the implementation of the rule-based emission calculator that applies various adjustment factors to ensure realistic results.

This part of the implementation defines VayuNetra's rule-based emission calculator. The TripInput data class collects trip inputs such as mode, fuel, distance, occupancy, vehicle age, proximity to industrial zones, and base emission factors (in gCO_2/km). Helper functions apply an age penalty (+1% per year after 3 years, capped at 15%), an industrial context multiplier (+5% if near industrial zones), and a carpool sharing divisor (based on occupancy count). The core computation then combines these adjustments to calculate and return the per-person emission in $kgCO_2$.

This rule-based approach allows the system to dynamically adapt to user inputs, ensuring that emissions reflect real-world variables like vehicle wear, travel conditions, and industrial influence on air quality.

```
# app/management/commands/seed_factors.py
from django.core.management.base import BaseCommand
from app.models import EmissionFactor

DEFAULTS = [
    ("car", "petrol", 192.0),
    ("car", "diesel", 171.0),
    ("carpool", "petrol", 192.0),  # shared later via occupancy
    ("bus", "diesel", 82.0),
    ("train", "electric", 25.0),
    ("bike", "na", 0.0),
    ("walk", "na", 0.0),
]

class Command(BaseCommand):
    help = "Seed default emission factors (gCO2/km per person)."

    def handle(self, *args, **kwargs):
        for mode, fuel, gpkm in DEFAULTS:
            obj, _ = EmissionFactor.objects.update_or_create(
                mode=mode, fuel=fuel, defaults={"factor_g_per_km": gpkm}
            )
        self.stdout.write(self.style.SUCCESS("Seeded emission factors."))
```

Figure 4.3: Seeding Default Emission Factors for Rule-based Calculation

Before any emission calculations can be performed, the system requires a set of default emission factors to serve as reference values for each mode and fuel combination. The following snippet demonstrates how this initialization process is automated using a Django management command.

This section seeds VayuNetra's database with default emission factors (gCO_2/km per person) for common mode–fuel pairs using a custom Django command. The predefined list includes baseline values such as *Car–Petrol (192.0)*, *Car–Diesel (171.0)*, and *Train–Electric (25.0)*. The script ensures that each factor is either inserted or updated idempotently, maintaining data consistency across deployments. Executing this command initializes the emission calculation module with standardized baseline data, eliminating the need for manual configuration.

With these seeded factors, VayuNetra ensures that every emission calculation is based on accurate and uniform reference data, forming the cornerstone of its rule-based computation framework.

4.2 System Access and Usage Procedure

This section explains the step-by-step process for accessing and interacting with the VayuNetra: AI-Powered Carbon Footprint and Emission Tracking System. Users can register or log in through the web platform to access personalized dashboards. Once logged in, they can record daily travel activities, enter transportation details such as mode, fuel type, and distance, and automatically calculate the corresponding carbon emissions. The system processes this data through predefined emission factors and algorithms to generate accurate carbon credit values. Users can then view their credits, monitor their CO_2 savings, and participate in carbon credit trading within the platform.

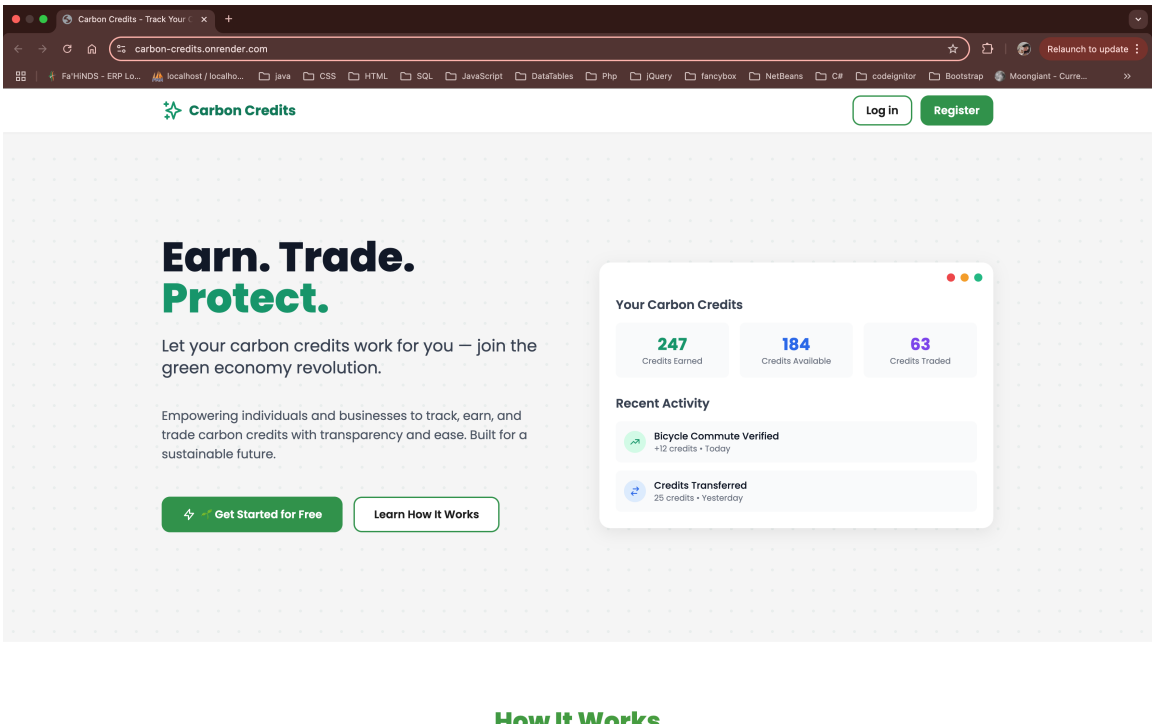

Figure 4.4: VayuNetra Homepage Interface

Figure 4.4 illustrates the homepage interface of the VayuNetra system. The page serves as the primary access point for all users, providing options for Login and Register at the top-right corner to facilitate secure user authentication. The navigation bar includes links to key sections such as Home, Features, and About, allowing users to explore the platform easily. The central section highlights the purpose of the system — to enable users to track, earn, and trade carbon credits — through clear headings and action buttons like Get Started and Learn More. On the right side of the page, a summary card displays user-related information such as Credits Earned, Credits Available, and Credits Traded, along with a Recent Activity section that provides quick updates on recent transactions. Overall, the homepage presents a well-structured and user-friendly layout that guides users toward understanding the platform's objectives and accessing its main functionalities.

Figure 4.5 shows the Employee Dashboard of the VayuNetra system. This page allows employees to track their sustainable commuting activities, monitor total carbon credits earned, and view their environmental impact.

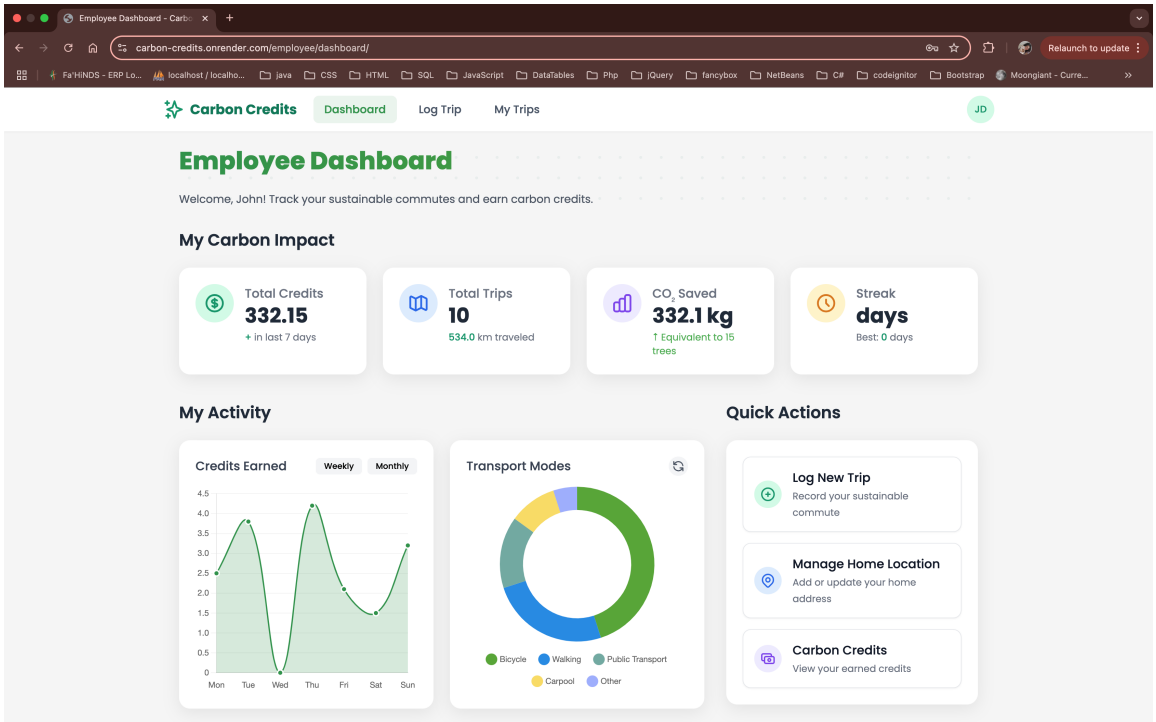

Figure 4.5: Employee Dashboard Interface

The top section displays key performance indicators such as Total Credits, Total Trips, CO_2 Saved, and daily Streaks. Below that, the My Activity section presents analytical charts—showing weekly credit trends and transport mode distribution—to help users visualize their contribution patterns. The right side includes a Quick Actions panel with options to Log New Trip, Manage Home Location, and view Carbon Credits. Overall, this dashboard offers a clean, data-driven summary of an employee's eco-friendly activities and provides quick navigation for further engagement in emission tracking.

Figure 4.6 illustrates the Employer Dashboard of the VayuNetra platform. This interface allows organizations to monitor and manage their company-wide carbon footprint through aggregated employee data.

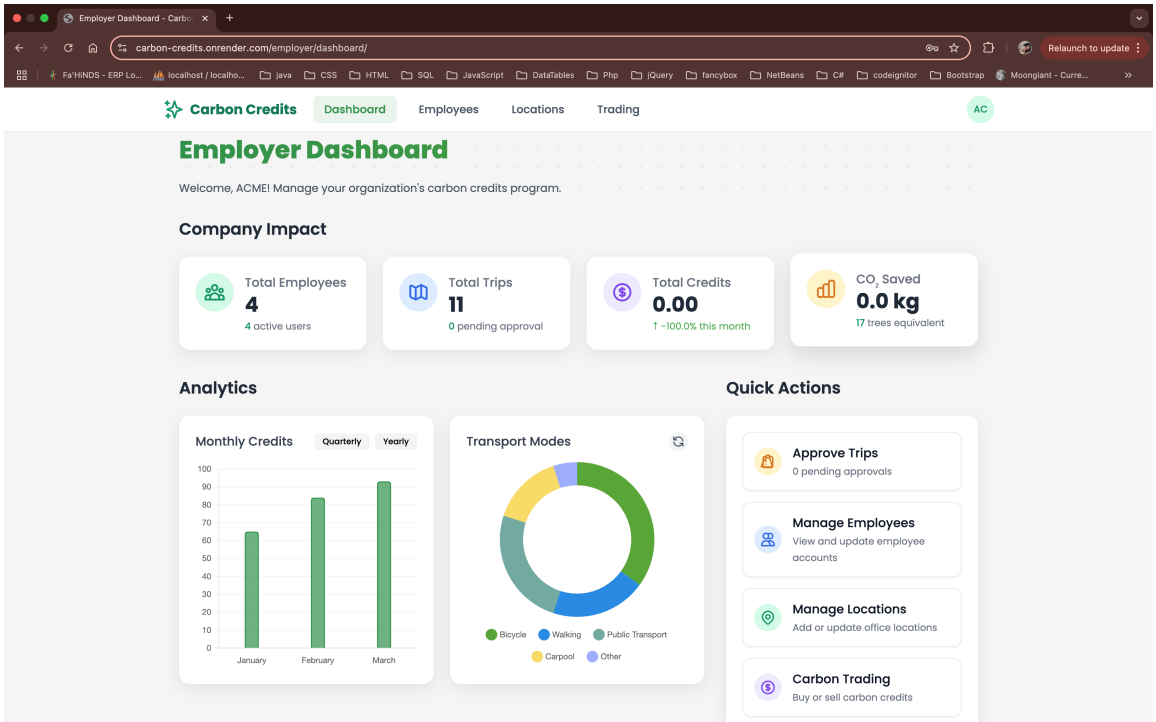

Figure 4.6: Employer Dashboard Interface

The top section presents Company Impact metrics, including the total number of employees, total trips recorded, total credits earned, and total CO_2 saved. These insights help employers evaluate the organization's overall sustainability performance. The Analytics section provides a visual overview through bar and pie charts, displaying monthly credit trends and transport mode distribution among employees. On the right-hand side, the Quick Actions panel enables employers to perform essential management tasks such as approving trips, managing employee accounts, updating office locations, and overseeing carbon credit trading. Overall, this dashboard serves as a centralized control panel for employers to track, approve, and enhance their organization's environmental contributions.

Figure 4.7 displays the Super Admin Dashboard of the VayuNetra platform. This interface provides administrators with complete oversight of the system's operations, including user activity, organizational performance, and overall platform metrics.

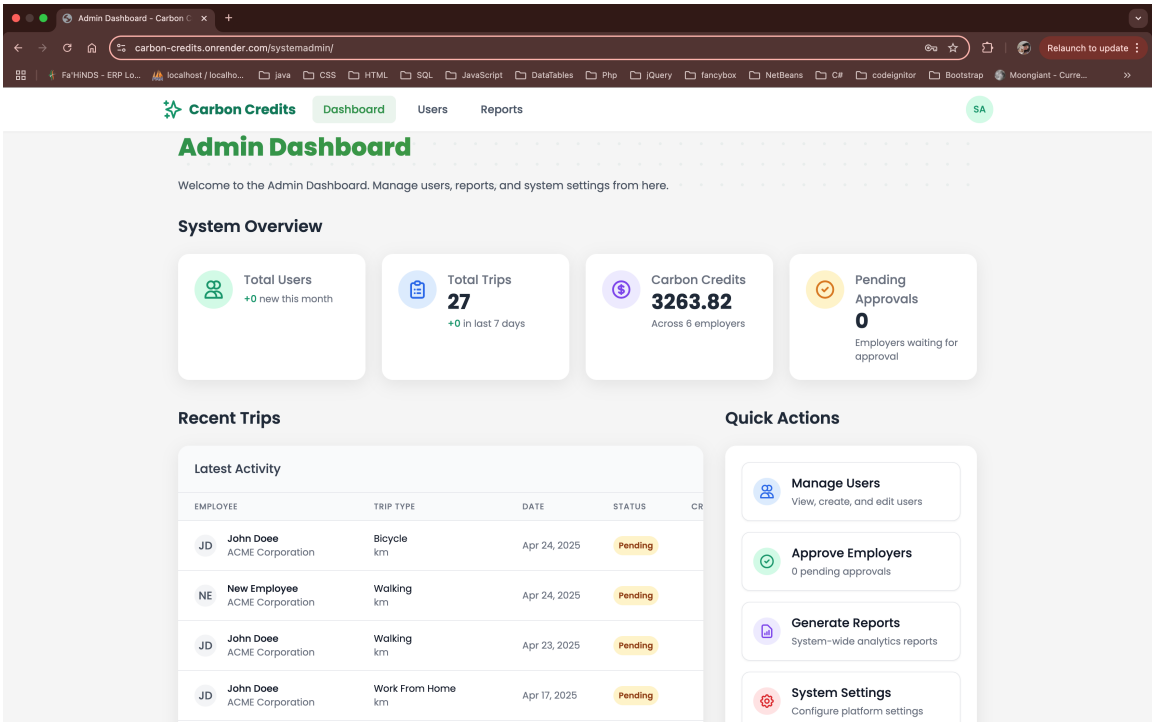

Figure 4.7: Super Admin Dashboard Interface

The System Overview section at the top summarizes total users, total trips logged, total carbon credits generated, and pending approval requests. The Recent Trips table below lists the latest commuting activities submitted by users along with their approval status and timestamps. On the right side, the Quick Actions panel allows administrators to manage users, approve new employers, generate system-wide reports, and modify platform settings. This dashboard ensures transparent monitoring, efficient management, and smooth operation of the entire VayuNetra ecosystem.

4.3 Timeline - Semester VII

This section presents the timeline of the VayuNetra: AI-Powered Carbon Footprint and Emission Intelligence Platform project, highlighting the major milestones achieved during Semester VII. The execution plan is illustrated through a Gantt Chart (Figure 4.8), which visually represents the project's key phases, dependencies, and progress achieved so far. The timeline demonstrates how the project evolved from initial conception to partial implementation, with remaining modules planned for completion in the upcoming development cycle.

The project is divided into two main phases: Phase One – Project Conception and Initiation, and Phase Two – System Design and Implementation. This structured approach ensured a clear progression path, efficient coordination among team members, and alignment with the project's objectives of sustainability and intelligent emission management.

4.3.1 Phase One: Project Conception and Initiation

This phase focused on establishing the foundation for the system. Major tasks included identifying the real-world problem of transportation-based emissions, reviewing existing carbon tracking systems, and defining the system's objectives. The literature review and feasibility study helped identify gaps in existing solutions and shaped the development of VayuNetra's unique rule-based emission model. The team also finalized the project architecture, selected the technology stack (Django, HTML, Tailwind CSS, SQLite), and designed early mockups of the user interface to visualize user interactions. This stage laid the groundwork for the platform's functional and technical development.

4.3.2 Phase Two: System Design and Implementation

he second phase focused on the development and implementation of VayuNetra's core modules. At this stage, two primary objectives have been successfully achieved:

- Carbon Footprint Calculation: Implementation of a rule-based model to calculate user-specific emissions based on distance, fuel type, travel mode, vehicle condition, and occupancy.
- AI Sustainability Insights: Integration of a natural language processing module that provides personalized eco-friendly recommendations and awareness tips based on travel data.

The upcoming development cycle will focus on integrating the remaining modules:

- Carbon Credit and Wallet System: To reward users and organizations for eco-friendly practices through a secure digital credit system.
- Community Dashboards: To display aggregated data, leaderboards, and emission trends for collaborative sustainability tracking.

Once completed, these additions will make VayuNetra a holistic platform for intelligent emission tracking, behavioral motivation, and collective climate action.

4.3.3 Progress Overview

The progress achieved so far reflects structured development, continuous evaluation, and a systematic approach to meeting the project's objectives. Each milestone was executed according to a defined timeline, ensuring that both technical and functional goals were progressively realized throughout the implementation cycle. The development process has been divided into well-defined stages covering system design, backend implementation, AI integration, and frontend development.

The following key milestones highlight the progress achieved during the project lifecycle:

- **Completion of Planning and Literature Review:** An extensive literature review and project planning phase were completed, helping to identify existing research gaps in carbon tracking systems and define the core objectives and scope of VayuNetra. This stage laid the conceptual foundation for the system's AI-powered emission tracking model.

- **Architectural Design and Database Modeling:** A detailed system architecture was designed to define data flow between modules, including emission factor storage, trip logging, credit management, and user authentication. The database schema was implemented in SQLite, ensuring reliable data handling and modular scalability.

- **Implementation of the Rule-Based Emission Calculator and AI Insights:** The emission calculation engine was successfully developed using rule-based formulas to compute CO_2 emissions per trip, incorporating contextual adjustments such as fuel type, occupancy, and vehicle age. In addition, an AI-based insights module was integrated to provide personalized sustainability recommendations and behavior-driven suggestions.

- **User Authentication and Role-Based Access Control:** A secure authentication mechanism was implemented to manage different user roles — including employees, employers, banks, and administrators. Role-based access ensures that each user interacts only with authorized components of the system, maintaining both data privacy and operational integrity.

- **Frontend Development and Visualization:** User-friendly web interfaces were developed using Django and Tailwind CSS. Core pages such as *Trip Logging*, *Emission Tracking*, and *Dashboard Visualization* were implemented, providing real-time summaries of emission data, carbon credits, and sustainability performance.

- **Upcoming Enhancements:** The next stage of development focuses on implementing the Carbon Credit System, which converts emission savings into digital rewards, and the Community Dashboard, which visualizes collective emission data across organizations. Additionally, performance optimization, advanced analytics, and testing modules are scheduled for the upcoming iteration.

Regular monitoring and version control through GitHub have ensured that all development milestones are tracked effectively, with each update documented and tested before deployment. Weekly reviews were conducted to assess technical progress, fix implementation issues, and align tasks with project objectives.

The **Gantt Chart** shown below provides a visual representation of the project's timeline, highlighting completed milestones, ongoing tasks, and upcoming deliverables. It demonstrates that the project is on track toward its overarching vision — enabling smarter, data-driven, and sustainable transportation through AI-assisted emission intelligence.

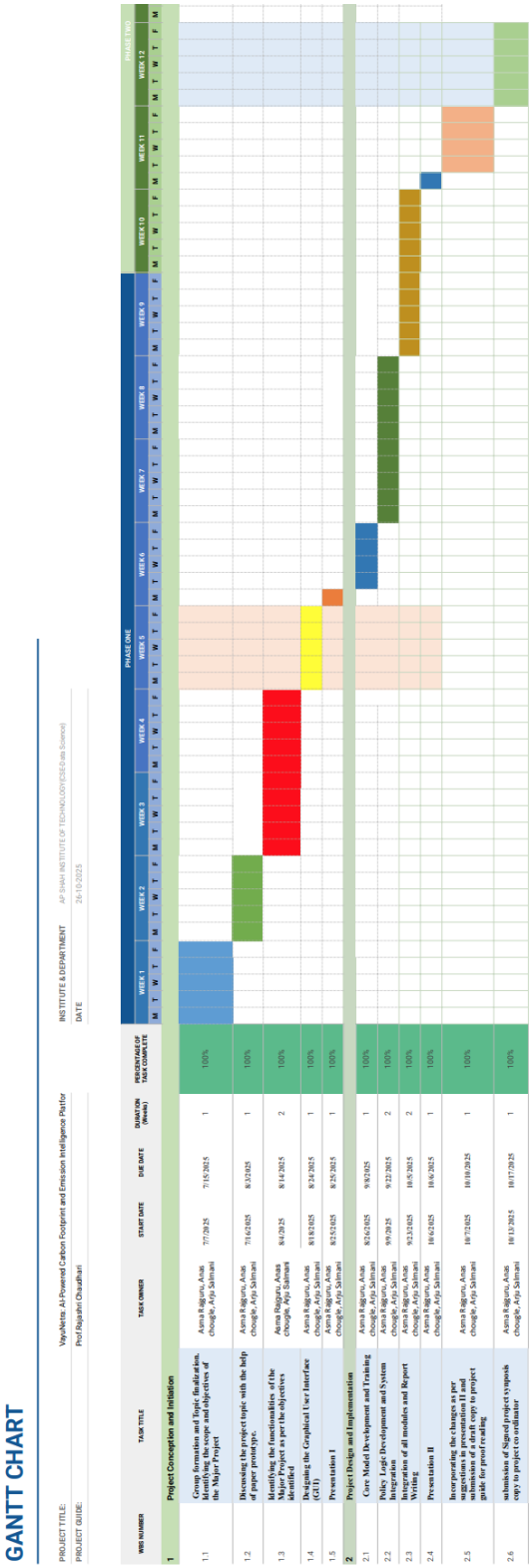

Figure 4.8: Gantt Chart showing the current project progress

Chapter 5

Testing

5.1 Software Testing

This section describes the test plan, environment, and methods adopted to validate the core features of VayuNetra: AI-Powered Carbon Footprint and Emission Intelligence System. Testing focused on correctness of rule-based emission calculations, reliability of data flow (models, services, views), integrity of the carbon-credit logic, and usability of role-based dashboards (employee, employer, bank, admin).

Test Environment

- Backend: Django (Python 3.x), SQLite (development & testing).
- Frontend: HTML + Tailwind CSS (Chrome, Firefox).
- Deployment Target: Render (staging), local dev server for tests.
- Libraries/Tools: pytest/unittest (unit tests), Django Test Client (integration), Browser DevTools (UI checks), coverage.py (code coverage).

Testing Objectives

- Verify emission computation correctness for multiple transport modes, fuels, distances, and edge conditions (vehicle age, occupancy, industrial proximity).
- Ensure data consistency across EmissionFactor, Trip, and EmissionRecord.
- Validate role-based access (employees, employers, banks, admins) and authorization boundaries.
- Confirm carbon credit generation and wallet ledgers reflect computed emissions and adjustments.
- Assess UI responsiveness and clarity of dashboards, charts, and summaries.

- Check basic security (auth flows, input validation, server-side checks) and error handling.

Testing Methodology

Unit Testing: Pure functions (e.g., age penalty, industrial multiplier, per-person divisor) and the core calculator were validated with deterministic inputs and expected outputs (including boundary cases like negative/zero distance blocked at validation).

Integration Testing: End-to-end flows using the Django Test Client: create a user, log a trip, compute emission, persist EmissionRecord, update credit wallet render dashboard cards and charts.

Functional/UI Testing: Manual checks of forms, tables, and charts for each role. Verified navigation, filters, and error messages. Cross-browser checks on common viewports.

Performance/Resilience (basic): Measured API response time for trip submission and dashboard rendering under small batches (10–100 trips). Verified graceful handling of invalid inputs and empty datasets.

Sample Calculation Validation

Given: Car–Petrol factor = $192\,\mathrm{gCO_2/km}$, distance = 10 km, vehicle age = 5 years, not near industrial zone, occupancy = 1.
Age penalty = $\min((5-3)\times 1\%, 15\%) = 2\%$.
Base = $10 \times 192/1000 = 1.92$ $\mathrm{kgCO_2}$. Adjusted = $1.92 \times 1.02 = 1.9584$ $\mathrm{kgCO_2}$. Expected stored value ≈ 1.958 $\mathrm{kgCO_2}$ (rounded).

5.2 Functional Testing

This section enumerates feature-level tests executed to verify that each core function meets its acceptance criteria. Test data uses seeded emission factors and realistic trip inputs.

Table 5.1: Functional Testing of Core Modules in VayuNetra

Feature	**Test Case & Steps**	**Expected Result**
User Authentication	Register new user → ensure unique username/email → login → session created.	Unique credentials validated; secure login; redirect to appropriate dashboard.
Role-Based Access	Login as employee, employer, bank, and admin; attempt access to restricted pages.	Authorized views load successfully; restricted routes return 403 or redirect.
Trip Logging (valid)	Submit trip: mode=car, fuel=petrol, distance=12.5 km, age=6, occupancy=1, near_industrial=false.	Trip stored correctly; validation passed; confirmation message displayed.
Trip Logging (invalid)	Enter invalid data: distance 0, non-numeric input, or invalid mode/fuel pair (e.g., walk + diesel).	Validation error shown; trip not stored in database.
Emission Calculation	Create trip → compute emission (rule-based) → generate `EmissionRecord`.	Emission ($kgCO_2$) matches formula; result stored and displayed accurately.
Carpool Sharing	Mode = carpool, distance = 20 km, occupancy = 4. Verify per-person division of emissions.	Emission correctly divided by occupancy; consistent rounding observed.
Vehicle Age Penalty	Compare vehicle age = 2 vs 8 years.	Correct multiplier applied (0% vs capped at 15%); stored result accurate.
Industrial Context	Set near_industrial = true and compute emission.	5% increase applied; contextual note visible in record.
Credit Generation	New `EmissionRecord` triggers credit computation and ledger entry.	Credits generated correctly; wallet totals updated automatically.
Employee Dashboard	View metrics (Total Trips, Credits, CO_2 Saved), charts (weekly trends, mode share).	Accurate data displayed; charts render without errors.
Admin Oversight	Review users, employers, emission factors, and trip logs.	Data updates persist; actions logged for audit trail.

Representative Functional Test (Calculation Path)

Input: mode=carpool, fuel=petrol, distance=24 km, occupancy=3, age=7 years, near_industrial=true.
Base factor = 192 g/km $\Rightarrow$ base = $24 \times 0.192 = 4.608$ kg.
Age penalty = $(7 - 3) \times 1\% = 4\% \Rightarrow 1.04$. Industrial = 1.05. Per-person divisor = 3.
Adjusted = $(4.608 \times 1.04 \times 1.05) \div 3 = 1.667$ kgCO_2 per person (rounded).
Expected: EmissionRecord.kg_co2 ≈ 1.667; dashboard aggregates update; credit logic triggered.

Acceptance Criteria

- All unit tests for calculator helpers and main compute function pass with $\geq 95\%$ coverage for the calculation module.
- Trip submission, emission record creation, and credit ledger entry succeed atomically or roll back on failure.
- Role-based pages enforce correct authorization; no privilege escalation paths.
- Dashboards display accurate aggregates and handle empty or large datasets gracefully.
- Input validation prevents nonsensical data (negative distance, invalid mode/fuel combinations).

Chapter 6

Result and Discussions

This chapter presents the results and evaluation of the VayuNetra: AI-Powered Carbon Footprint and Emission Tracking System. The system was tested to assess how accurately it calculates transportation-based carbon emissions, how consistently it generates and manages carbon credits, and how efficiently it processes user data across different modules. Using Django for backend processing and a rule-based AI emission model, the platform successfully produced reliable and meaningful outputs for both individual users and organizations.

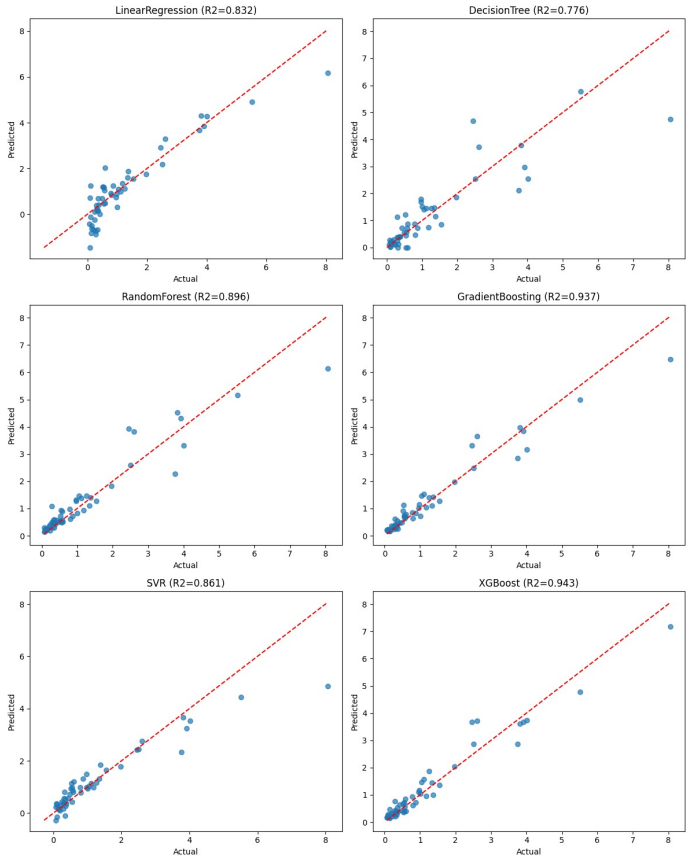

Figure 6.1: Model Performance Comparison using Regression Algorithms (R^2 Scores)

In Figure 6.1 presents a comparison of six regression models—Linear Regression, Decision Tree, Random Forest, Gradient Boosting, Support Vector Regression (SVR), and XGBoost—used for emission prediction within the VayuNetra system. Each subplot shows the relationship between the actual and predicted carbon emission values, where points closer to the red dashed line ($y = x$) indicate higher prediction accuracy.

Among the tested models, XGBoost achieved the highest performance with an $\mathbf{R^2}$ score of 0.943, followed by Gradient Boosting (0.937) and Random Forest (0.896). These ensemble-based models effectively captured the non-linear relationships between travel parameters and emissions, resulting in more accurate predictions. The Linear Regression and Decision Tree models, while simpler, showed relatively lower accuracies ($\mathbf{R^2} = 0.832$ and 0.776 respectively) due to their limited capacity to handle complex dependencies. SVR demonstrated moderate performance ($\mathbf{R^2} = 0.861$), balancing bias and variance well.

Overall, the results indicate that ensemble learning methods—particularly XGBoost and Gradient Boosting—are best suited for VayuNetra's emission estimation task, providing reliable and consistent results across diverse travel and contextual inputs.

To benchmark the regression estimators used for VayuNetra's CO_2 emission prediction, we compare six models using two complementary metrics: the coefficient of determination (R^2) to assess goodness of fit, and the Root Mean Squared Error (RMSE) to quantify average prediction error (lower is better). The following figures summarise these results.

Model fit by R^2: Higher bars indicate better alignment between predicted and observed emissions.

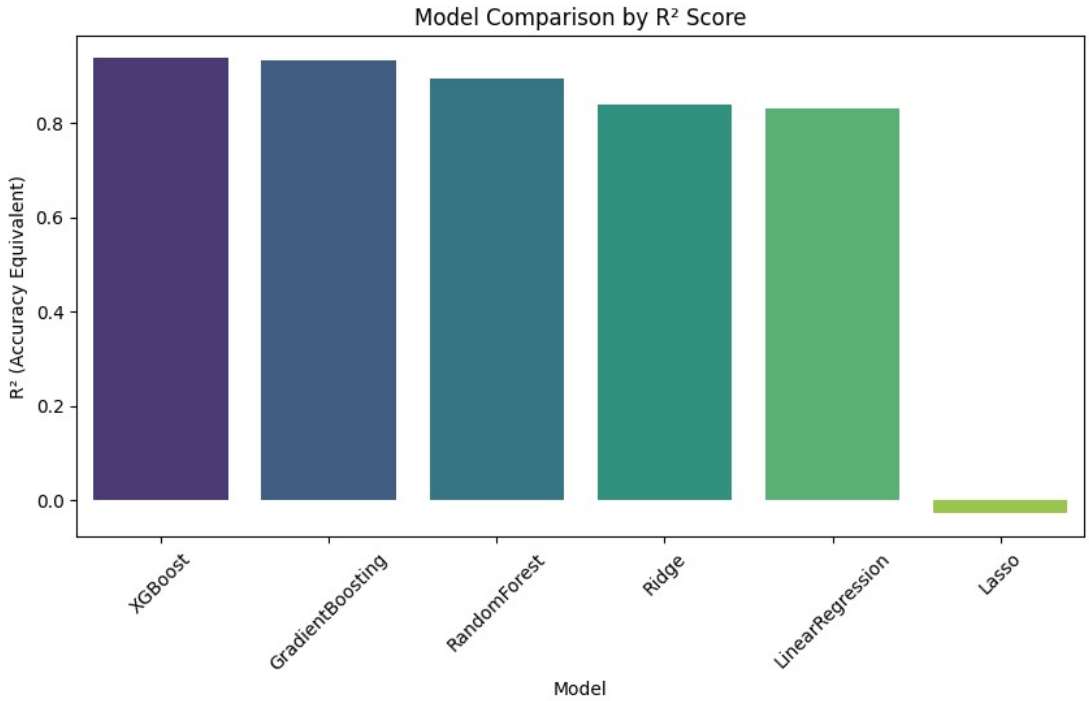

Figure 6.2: Model Comparison by R^2 Score

Boosting-based ensembles (XGBoost, GradientBoosting) achieve the strongest fit ($R^2 \approx 0.92$–0.94), followed by RandomForest. Linear/Ridge trail slightly, while Lasso underfits this problem.

Prediction error by RMSE: Lower bars indicate smaller average deviations from ground truth.

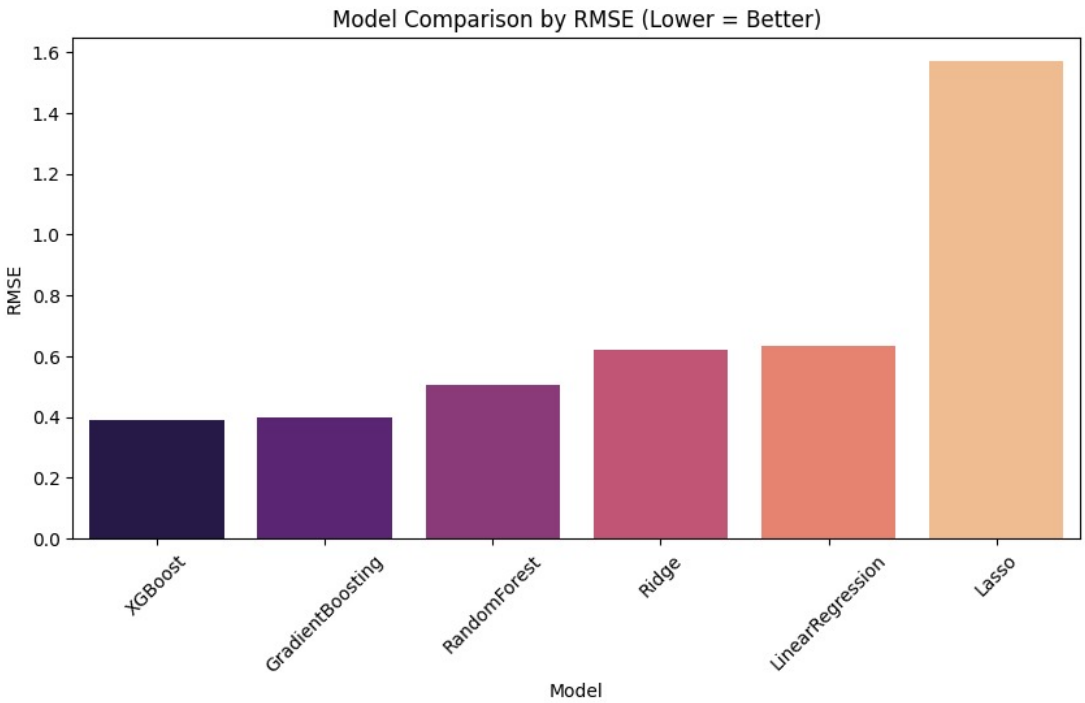

Figure 6.3: Model Comparison by RMSE (Lower = Better)

XGBoost and GradientBoosting yield the lowest errors (RMSE ≈ 0.39–0.40), confirming their stability; RandomForest and Ridge are competitive; Lasso exhibits the highest error and is not recommended for deployment.

Overall, these results support prioritising a boosting-based ensemble for VayuNetra's predictive layer, while retaining the rule-based calculator for transparency and fast, interpretable baselines.

In Figure 6.4 shows the Predicted vs Actual (Binned Heatmap) for the emission estimation results generated by VayuNetra's regression model. In this plot, both predicted and actual emission values are grouped into discrete bins for easier visual comparison. Each cell in the matrix represents the number of predictions that fall into a particular bin pair, with darker shades indicating higher counts.

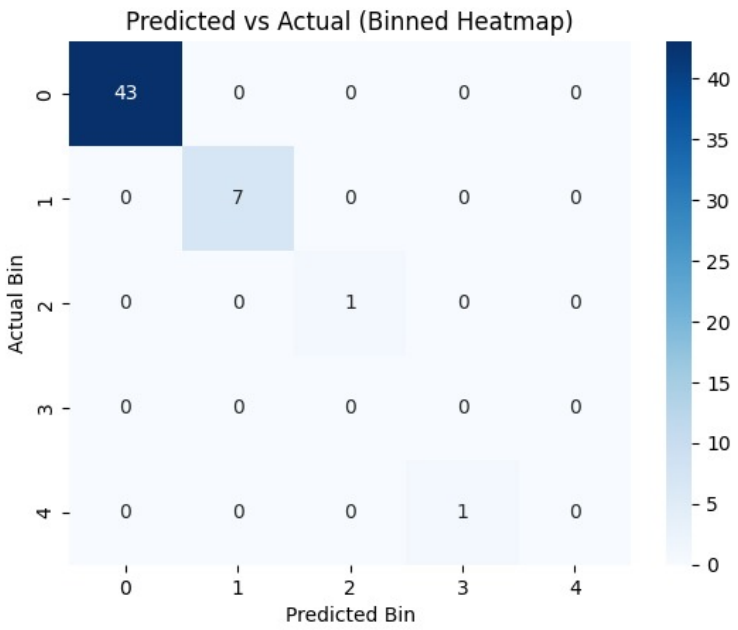

Figure 6.4: Predicted vs Actual Emission Values (Binned Heatmap Representation)

As observed, the majority of values are concentrated along the diagonal cells, especially in the top-left region, which corresponds to lower emission bins. This indicates that most predictions closely match the actual observed emissions, confirming the model's high reliability for low-to-moderate emission ranges—where most real-world trips typically fall. Only a few instances appear off the diagonal, reflecting minor deviations in higher bins.

Overall, the heatmap demonstrates strong alignment between predicted and actual outputs, validating the model's ability to accurately estimate trip-level CO_2 emissions. This reinforces VayuNetra's effectiveness in providing realistic and trustworthy emission predictions across different transportation modes.

Chapter 7

Conclusion

The *VayuNetra* project introduces an innovative, AI-driven approach to environmental sustainability by focusing on transportation-based carbon footprint tracking and reduction. It bridges a critical gap between individual awareness, data transparency, and collective environmental action. By combining rule-based emission calculations, real-time sustainability insights, and a digital carbon credit system, *VayuNetra* transforms the complex process of carbon accounting into an accessible, engaging, and motivating experience for users. The platform enables individuals and organizations to understand the environmental impact of their daily travel choices while promoting responsible behavior through actionable feedback and gamified incentives.

Through the integration of AI, geolocation data, and intelligent analytics, *VayuNetra* ensures accurate and context-aware emission assessments. It empowers users to make greener decisions by providing personalized eco-tips and transparent tracking of their carbon savings. The inclusion of role-based management for employees, employers, and administrators further enhances accountability, enabling a structured framework for large-scale participation in emission reduction efforts. This makes the system adaptable for various settings, including workplaces, schools, and communities, encouraging collaboration toward a common goal of sustainable mobility.

By addressing the limitations of existing carbon tracking tools—such as lack of personalization, real-time insights, and reward mechanisms—*VayuNetra* demonstrates how technology can inspire meaningful behavioral change. Its user-friendly interface, data-driven insights, and credit-based reward system collectively foster long-term commitment to environmental responsibility.

Looking ahead, *VayuNetra* has the potential to evolve into a comprehensive sustainability intelligence platform. Future versions may incorporate IoT data, blockchain-based credit verification, and expanded APIs for integration with government and corporate carbon initiatives. With continuous enhancement and large-scale deployment, *VayuNetra* can play a vital role in supporting India's climate goals under the Paris Agreement and contribute significantly toward achieving a low-carbon, sustainable future.

Chapter 8

Future Scope

The future scope of VayuNetra lies in expanding its reach and enhancing its technological capabilities to make carbon tracking more accurate, automated, and impactful. The platform can be further developed by integrating IoT-based vehicle sensors, GPS data, and public transport APIs to enable real-time emission monitoring without manual data entry. Advanced AI and machine learning models can be introduced to predict future emission patterns, provide personalized recommendations, and analyze behavioral trends. To ensure greater transparency and security, a blockchain-based carbon credit mechanism can be implemented, allowing verified credit generation and trading among users, organizations, and governing bodies.

Additionally, VayuNetra can be extended beyond individuals and organizations to include municipal bodies, schools, and logistics sectors, enabling large-scale community emission tracking and collective sustainability programs. The development of a mobile application with gamified features such as streaks, leaderboards, and challenges would encourage long-term user engagement. The system can also be aligned with government initiatives and corporate CSR programs, allowing users and companies to earn officially recognized carbon credits. With continuous improvements in data visualization, predictive analytics, and educational outreach, VayuNetra has the potential to evolve into a comprehensive sustainability intelligence platform that contributes significantly to India's Net-Zero 2070 mission and inspires people to adopt environmentally responsible lifestyles globally.

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